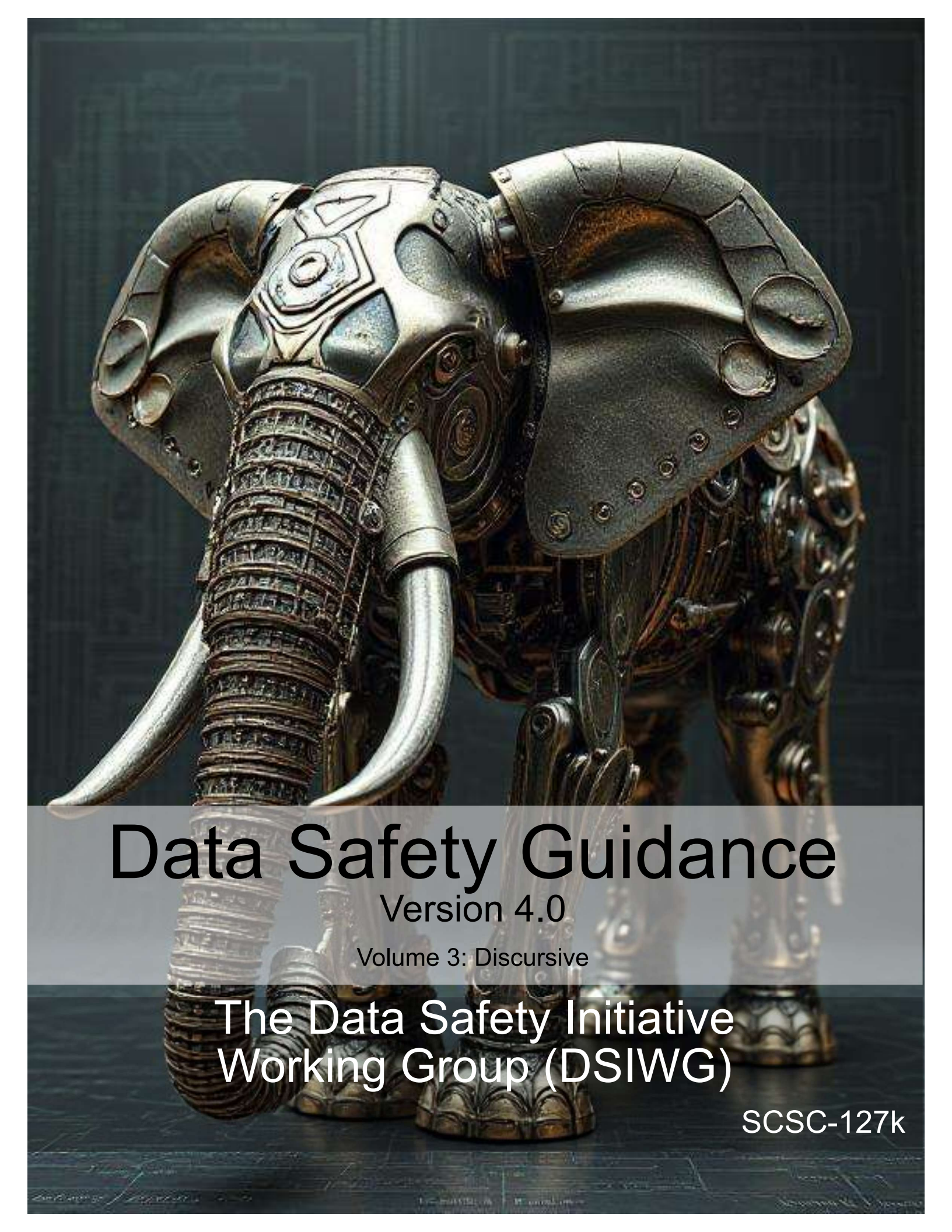




Data Safety Guidance

Version 4.0

Volume 3: Discursive

The Data Safety Initiative
Working Group (DSIWG)

SCSC-127k

SCSC Publication Number: SCSC-127J

This work is licensed under the Creative Commons Attribution 4.0 International License. To view a copy of this license, visit <http://creativecommons.org/licenses/by/4.0/> or send a letter to Creative Commons, PO Box 1866, Mountain View, CA 94042, USA. You are free to share the material in any form and adapt the material for any purpose providing you attribute the material to the [Safety-Critical Systems Club \(SCSC\) Data Safety Initiative Working Group](#), reference the source material, include the licence details above, and indicate if any changes were made. See the license for full details.

This document was prepared using the $\text{\LaTeX}2_{\epsilon}$ typesetting system.

Editing and typesetting by Mark Templeton and Tim Rowe.

Cover design by DeepAI.org and Tim Rowe, based on an original by Paul Hampton.

The [Safety-Critical Systems Club \(SCSC\)](#) is the professional network for sharing knowledge regarding safety-critical systems. It brings together:

- engineers and specialists from a range of disciplines working on safety-critical systems in a wide variety of industries;
- academics researching the arena of safety-critical systems;
- providers of the tools and services that are needed to develop the systems; and
- the regulators who oversee safety.

Through publications, seminars, workshops, tutorials, a web site and, most importantly, at the annual [Safety-critical Systems Symposium \(SSS\)](#), it provides opportunities for these people to network and benefit from each other's experience in working hard at the accidents that don't happen. It focuses on current and emerging practices in safety engineering, software engineering, and product and process safety standards.

This document was written by the [Data Safety Initiative Working Group \(DSIWG\)](#), which is convened under the auspices of the [SCSC](#). The document supports the [DSIWG's](#) vision, which is to have clear guidance that reflects emerging best practice on how data (as distinct from software and hardware) should be managed in a safety-related context. This update takes account of the consensus that a process-based guidance document will complement existing safety management processes, making it more usable. It was formally released at [SSS'25](#), 4–6 February 2025 details of which may be found at <https://scsc.uk/e1099>

Comments on this document are actively encouraged. These can be emailed to:

comments@data-safety.scsc.uk.

Alternatively, a comments submission form is available at:

data-safety.scsc.uk/comments.

While the authors and the publishers have used reasonable endeavours to ensure that the information and guidance given in this work is correct, all parties must rely on their own skill and judgement when making use of this work and obtain professional or specialist advice before taking, or refraining from, any action on the basis of the content of this work. Neither the authors nor the publishers make any representations or warranties of any kind, express or implied, about the completeness, accuracy, reliability, suitability or availability with respect to such information and guidance for any purpose, and they will not be liable for any loss or damage including without limitation, indirect or consequential loss or damage, or any loss or damage whatsoever (including as a result of negligence) arising out of, or in connection with, the use of this work. The views and opinions expressed in this publication are those of the authors and do not necessarily reflect those of their employers, the [SCSC](#) or other organizations.

Data Safety Guidance

The Data Safety Initiative Working Group [DSIWG]

February 2025

Change History

Version	By	Status	Date
1.0	The DSIWG Team	First draft for external review	31-JAN-2014
1.1	The DSIWG Team	(Internal edition for DSIWG use only)	09-DEC-2014
1.2	The DSIWG Team	For publication at SSS '15	23-JAN-2015
1.3	The DSIWG Team	For publication at SSS '16	29-JAN-2016
2.0	The DSIWG Team	For publication at SSS '17	30-JAN-2017
3.0	The DSIWG Team	For publication at SSS '18	26-JAN-2018
3.1	The DSIWG Team	For publication at SSS '19	01-FEB-2019
3.2	The DSIWG Team	For publication at SSS '20	11-FEB-2020
3.3	The DSIWG Team	For publication at SSS '21	09-FEB-2021
3.4	The DSIWG Team	For publication at SSS '22	08-FEB-2022
3.5	The DSIWG Team	For publication at SSS '23	07-FEB-2023
3.6	The DSIWG Team	For publication at SSS '24	13-FEB-2024
3.7	The DSIWG Team	For publication at SSS '25	04-FEB-2025
4.0	The DSIWG Team	For publication at SSS '26	10-FEB-2026

Changes Since the Last Edition

The main changes in this edition are:

- the inclusion of a process flow diagram at Section [??](#);
- the inclusion of an appendix (Appendix [??](#)) on [AI](#) and autonomy; and
- the inclusion of an appendix ([Appendix 3](#)) on the RADISH tool.

The inclusion of the new Appendix [??](#) and [Appendix 3](#) means that subsequent appendix numbers have changed.

Hyperlinks (in the electronic edition) to abbreviations, acronyms and defined terms have been substantially expanded and further abbreviations, acronyms and defined terms have been added.

The body of text the guidance, but not yet the appendices, has been edited for clarity and consistency.

Discursive definitions of terms and abbreviations not used in the document have been removed, and new discursive definitions and definitions have been added.

The definition of hazard has been expanded to include harm, not just accidents.

An inconsistency in the worked example regarding whether there is a contract for the system has been corrected.

To assist users of earlier 3.x versions of the guidance in ensuring that their existing data safety arguments have not been impacted by this update, a version of this document is available which has been annotated with change bars. To avoid clutter, minor changes that should not affect the meaning of the text have not been marked with change bars. The annotated version is available at <http://scsc.uk/scsc-127j>

Future work

MCA Ltd has continued to work with the [DSIWG](#) to develop a prototype software tool to assist in the automation of the processes described in this guidance document. A working version of the tool has been developed and organizations that could benefit from the use and further development of the tool are urged to contact MCA at [Mission Critical Applications Limited \(radish@mca-ltd.com\)](mailto:radish@mca-ltd.com).

A number of improvements to the guidance are currently planned. These improvements are intended to clarify the application of the data safety process and include:

- further detail on the assurance of communications and data flows,
- data safety considerations associated with distributed [datasets](#) and Blockchain,
- addition of new [treatments](#) to the tables in **??**,
- review of the tables of [treatments](#), with the aim of making them easier to use,
- further explanation of some [treatments](#), where their use or benefit is not immediately apparent,
- reordering of parts of the document to improve readability, especially as regards likelihood,
- further detail on tool assurance,
- harmonisation of language and guidance on how organizations may expand the tables to incorporate their own internal processes.
- guidance on the application of the data safety culture questionnaire,

Several of these changes are likely to cause parts of the document to be re-ordered – they have therefore been deferred to the next major update, in version 4.0 of the guidance.

If you or your organization are interested in learning more about the work of the [DSIWG](#) or joining any of the sub-groups, please visit the [SCSC](#) website, where more information including contact details may be found on the “Working groups” section of the site.

Related working groups

The [SCSC](#) sponsors initiatives to develop methods and techniques through a number of working groups. These groups each address safety aspects peculiar to their domain, including data aspects when appropriate. The current list of working groups includes:

- Assurance Cases,
- Security Informed Safety,
- Safe AI Working Group,
- Safer Complex Systems.

This page is intentionally blank

Foreword

Data is here. Data is growing. Data is causing harm.

Data is here: Data is becoming ever more important in our lives: influencing, managing and even controlling many critical aspects. The use of [artificial intelligence \(AI\)](#) systems is a new, exciting, but potentially hazardous use of data. [Large language model \(LLM\)](#) based systems are trained on vast amounts of data, and it is this data which enables them to be useful.

Some of this data is related to our personal safety and well-being. Consider, for example, the importance of data defining the layout of railway signals, data that indicates the position of underwater obstructions in nautical channels or data that is used to train a vision recognition system to detect tumours in medical images. Organizations now make significant decisions (including safety-related decisions) based solely on data held in systems. Hence, organizations need to safely manage, control and process their data. In particular, they must actively manage key [data properties](#) that preserve safety.

Data is growing: There are at least two reasons why the use of data has grown and, equally important, why it is expected to continue to grow. The first relates to the rapid expansion of the area loosely termed “Big Data”, including the use of large [datasets](#) to support machine learning and [AI](#) applications. The second is the growing use of systems of systems, where data is the lifeblood that connects together disparate elements and allows a cohesive capability to be built. Put simply, the need to address data-related issues is a pressing problem and will continue to be so.

Data is causing harm: Strictly speaking, data can neither cause nor prevent harm. However, mistakes in data, or the inappropriate use of data, within safety-related systems have been factors in a number of documented accidents and incidents. Examples include aircraft attempting to take off from the wrong runway (and consequently crashing), ships running aground, and patients being exposed to higher than planned doses of radiation.

Against this background, the [DSIWG](#) was established under the auspices of the [SCSC](#). The [DSIWG](#)’s aim is to develop clear, cross-sector guidance that reflects emerging best practice on how data (as opposed to software or hardware) should be managed in a safety-related context. For the most part, this guidance is based on well-established techniques, and it has been designed to be compatible with current safety standards and to integrate with existing safety management systems. What is new, however, is the explicit and relentless focus on data, making it a “first-class citizen” within system safety analyses. Because of this focus, this guidance should help organizations identify, analyse, evaluate and treat data-related risks, thus reducing the likelihood of data-related issues causing harm in the future.

Quick Start Guide

???consider whether the quick start guide needs to be included in all parts

Data really powers everything that we do.
Jeff Weiner

This section provides a single-page introduction to [data safety guidance \(DSG\)](#). For first-time readers this should help place individual sections within an appropriate context. It should also help returning readers quickly navigate the document's contents.

When reference is made to the content of other volumes, this is indicated by the volume number shown in bold text. So “Chapter **2:6**” refers to chapter 6 in volume 2.

Systems are changing. The role of data is becoming more prominent. Hence, data needs to be considered as a “first-class citizen” in system safety analyses. This will help mitigate organizational and system-level risks associated with the use of data.

The guidance is in three volumes, published as separate documents.

Volume 1 contains the normative material. It describes the “what” of the guidance and states what is needed to ensure conformance with this guidance, if required. It should be of use to those establishing processes and procedures for data safety assurance and for those doing the actual assurance;

Volume 2 contains the informative material. It describes the “how” of the guidance and should be of particular value to data safety practitioners;

Volume 3 (this volume) contains the discursive material. It describes the “why” of the guidance and gives background and supporting material intended to improve practitioner's understanding of the data safety management processes and of issues relating to data safety management.

A data safety management process has been developed. This is based on four phases:

- establish context;
- identify risks;
- analyse risks; and
- evaluate and [treat](#) risks.

???This part needs rewriting when structure is stable

- The underlying principles of the guidance are described in Chapter **1:2**.
- The objectives and outputs to satisfy the principles are described in Chapter **1:3**.
- Normative definitions and abbreviations are described in Chapter **1:4**.
- The activities of each phase (and associated [tailoring information](#)) are described in Chapter **??**.
- A worked example is provided in Chapter Chapter **2:??**.

- Guidance for the assignment of [data safety assurance levels \(DSALs\)](#) is given in Chapter **2:??**.
- Examples of ways that [DSALs](#) may be customised, with particular focus on likelihood are given in Chapter **2:??**.
- Issues relating to [AI](#) and [machine learning \(ML\)](#) are considered in Chapter **2:??** and Chapter **2:??**.
- Additional information on data categories is given in Chapter **2:13**.
- Issues that may arise when migrating, porting, importing or exporting data are discussed in Chapter **2:??**).
- Lifecycle considerations are addressed in Chapter **??**.
- The data safety tool RADISH is described in Chapter **??**.
- Some of the data issues that made management of the Covid-19 virus difficult are discussed in Chapter **??**.
- A summary of accidents and incidents in which data was potentially a causal factor is given in Chapter **??**.

We never seem to explain what we mean by data categories. Consider doing so.

This page is intentionally blank

Contents

1	Introduction	1
1.1	Aim and Scope	1
1.2	Intended Relationship to Other Documents	1
1.3	Normative, Informative and Discursive Text	2
1.4	Conformance	2
2	Lifecycle Considerations	3
2.1	Usage Scenarios	3
2.2	Ways that Data Can Cause Problems	3
2.3	Data Properties	6
2.3.1	The analysability property	7
2.3.2	The explainability property	7
2.4	Data in System Lifecycles	9
2.4.1	Tool Assurance	12
2.4.2	Test Data	13
2.4.3	Interfaces with Existing Assessments	14
3	The Data Safety Tool: RADISH (Discursive)	17
3.1	An introduction to the tool	17
3.2	A caveat	17
4	Covid-19	19
4.1	Covid-19 and Data	19
4.2	Systems Involved with Covid-19 Data	19
4.3	Falsification / Misinformation of Data	20
4.4	Rumsfeld's known unknown and unknown unknown data conundrum	20
4.5	Learning	21
5	Incidents and Accidents (Discursive)	23
5.1	General	23

5.2	Post Office Horizon System	29
5.3	Battle of Agincourt	31
5.4	Gemini V	31
5.5	What3Words	32
5.6	Java log4j Library Vulnerability	33
5.7	Immensa False Negative Covid-19 Tests	34
5.8	Covid-19 Test Results Silently Deleted by Excel	34
5.9	Boeing 737 MAX 8 Crashes	35
5.10	Loss of Soyuz-2.1b Rocket Carrying Meteor-M 2-1 Weather Satellite	36
5.11	Cambrian Line Data Loss	37
5.12	Loss of Irish Rescue Helicopter	38
5.13	Loss of Schiaparelli Mars Lander	38
5.14	Interception of Communications	39
5.15	A400M, Torque Calibration Parameters	40
5.16	RN Submarine, Trawler Karen	40
5.17	Turkish Airlines A330	41
5.18	Dallas Hospital Ebola Incident	42
5.19	Qantas Boeing 737 Take-Off	42
5.20	Qantas Boeing 737 Loading	43
5.21	Grounding of Navigator Scorpio	43
5.22	Loss of MQ-9 reaper	44
5.23	Boeing 737-33A at Chambery Airport, France	45
5.24	Loss of Hermes 450	45
5.25	Advocate Lutheran Hospital	46
5.26	Grounding of Sichem Osprey	47
5.27	Near Collision of Trains, Cootamundra	47
5.28	Cedars-Sinai Medical Center Scanner	47
5.29	Grounding of The Pride of Canterbury	48
5.30	LOT Flight 282	49

5.31 Annabella Container Ship – Baltic Sea	49
5.32 Comair Flight 5191	50
5.33 Überlingen Mid-Air Collision	51
5.34 Fort Drum Artillery Incident	51
5.35 Early Release from Washington State Prison	52
5.36 Mars Climate Orbiter	52
5.37 Crash into Nimitz Hill, Guam	53
5.38 San Bernardino Derailment and Pipeline Rupture	54
5.39 Lake Peigneur Drilling Accident	54
6 Acronyms, Definitions and Glossary (Discursive)	57
6.1 Acronyms	57
6.2 Definitions and Glossary	59
7 References	65
8 Acknowledgements	67
9 Contributors	69
Index of Locations	73
Index	75

This page is intentionally blank

List of Tables

2.1	Properties of data	7
2.2	Properties that can be lost through issues	8
4.1	Systems involving data used to manage the pandemic	20
5.1	Incidents and accidents (by domain)	24
5.2	Incidents and accidents	25

This page is intentionally blank

List of Figures

2.1	Consumer-focused integrity requirements	3
2.2	Development lifecycle	10
2.3	Operational lifecycle	11
2.4	Data supply chain	12
3.1	RADISH within the project environment	18

This page is intentionally blank

1 Introduction

???Consider whether the introduction needs to be included in all parts

We're entering a new world in which data may be more important than software.

Tim O'Reilly

1.1 Aim and Scope

This guidance document aims to:

- describe the data safety problem;
- provide methods for identifying and analysing levels of risk; and
- recommend methods and approaches for evaluating and treating those risks.

It has been written for a wide readership. Its target audience is all those who have an interest in or a responsibility for safety-related data within systems, including managers, developers, safety engineers, assurers (including independent safety auditors), regulators, and operators.

The document is also intended to cover a number of different sectors. It identifies a wide spectrum of safety-related data that exists in many forms within systems, from specification and requirements data to maintenance and disposal data, and everything in between. In particular, this document is not just concerned with numerical or well-structured data used during system operation.

While they are considered mature enough to be useful, the contents of the document represent current thoughts on what is a complex and evolving area. Furthermore, to allow it to be produced within a reasonable timescale, this edition focuses on key items. It is not intended to be exhaustive. For example, this guidance document does not consider issues relating to staff competence or organizational structure.

1.2 Intended Relationship to Other Documents

This document is intended to be used as a supplement to existing standards and norms that are relevant to the scope of the work being undertaken. It may be used to provide a deeper insight into the risks that data poses to the project team's outputs, allowing them to produce credible improvements to the safety argument. Where a standard or norm sets out specific data-related objectives then, unless agreed otherwise with the regulator or safety duty holder, they shall take precedence over the guidance provided herein.

In the longer term, the hope is that future standards and norms will take up relevant concepts, approaches and methods from those in this document. The [DSIWG](#) also hopes that organizations will include the concepts, approaches and methods in their own safety management processes.

1.3 Normative, Informative and Discursive Text

Three types of text are used within this guidance document:

Normative text, which is prescriptive. Typically, this text is restricted to describing objectives and outputs.

Informative text, which is descriptive text that is closely linked to the normative text. Typically, this text provides a suggested way by which conformance with the normative text may be achieved, but alternative means of conformance are possible.

Discursive text, which contains discussions that are relevant to the general topic of data safety, but which are not closely linked to the normative text. A discussion on the relationship between data and software is an example of such text. Descriptions of historical incidents and accidents are another.

Each section and appendix of this guidance document contains a single text type. The relevant type is indicated in the section or appendix title.

1.4 Conformance

There may be occasions when it is desirable or necessary to make a claim of conformance against the objectives listed in this document. Such a claim may be required, for example, if this document is explicitly included as a normative reference from a formal standard. Alternatively, it may be required as part of an organization's internal processes.

To facilitate conformance claims, the following terminology is used within the normative parts of this guidance document:

SHALL denotes items where evidence of conformance must be provided in order to claim conformance with this guidance document.

SHOULD denotes items where, in some circumstances, there may be valid reasons for not conforming with a particular item. The full implications of non-conforming must be understood, documented and approved in order to claim conforming with this guidance document.

MAY denotes items that are optional. These may be advantageous in some circumstances but not in others. Organizations are free to adopt any approach to these items without the need for further justification.

The terms have their normal English meanings in discursive and descriptive sections.

2 Lifecycle Considerations

Failure is an amazing data point that tells you which direction not to go.
Payal Kadakia

2.1 Usage Scenarios

If safety-related data is incorrect it can become dangerous when used, either by making a computer or control system perform inappropriate actions, or by misleading human users into making inappropriate decisions. Since the danger can only be determined when the usage of the data is understood, risk assessment should involve both the consumer and the producer of the data.



Figure 2.1: Consumer-focused integrity requirements

The producer investigates how the safety-related data is collected and what errors might occur. Building on activities in later phases of the data safety management process, the producer can provide some form of guarantee, or level of confidence, that the safety-related data meets the specific data-related requirements.

The consumer assesses the use of the safety-related data. In later phases of the data safety management process this [information](#) is used to define the required [data properties](#): for example, how accurate a particular safety-related [data artefact](#) must be.

In some cases a producer will be providing safety-related data without any knowledge of a specific user (e.g., mapping data or generic [databases](#) that are sold to many users). In these cases the producer will need to make some assumptions about possible users, and then clearly state what level of [integrity](#) the data has been produced to. It is then up to the users to check whether the declared [integrity](#) matches their need.

2.2 Ways that Data Can Cause Problems

???Originally said Volume 1, but it seems to fit here better. There are some risk-inducing issues that are different or more prevalent for data than for other system elements. An incomplete collection of examples is provided below. This list may provide a quick way of identifying risks, which could be especially useful at an early stage of a project:

Fluidity: Hardware and software can undergo significant amounts of product assurance and once assured may change relatively infrequently. Where change is required to hardware or software, it can be carefully managed and the impact on the safety case appraised. This is not always the case for data, which is often much more fluid. Indeed, the ease with which data can be changed is one motivation

for the move towards [data-driven systems](#). This fluidity means that it is not always possible to revisit safety cases when data changes. For example, the safety case for an autonomous vehicle cannot be updated every time that the vehicle acquires new knowledge during operation. Instead, the fact that data can change, along with any associated safety impacts, may need to be captured in the system safety case. Fluidity can also provide a temptation for unscrupulous operators to falsify data, for example, after an incident has occurred. Rigorous configuration control procedures can help protect against this type of behaviour.

Reuse: For the purposes of this discussion, “reuse” is interpreted as use of the same data in a different system or system context (e.g., lifecycle phase). Just because data was valid for use in a particular system, it does not immediately follow that it can be reused again in a similar system. Many considerations associated with data reuse are similar to those of software reuse, for example, similarity of requirements, similarity of role in the system, and similarity of required [integrity](#) / assurance level. One consideration that is different is that of [timeliness](#): data that was valid for use in a particular system at a particular time is not necessarily valid for reuse in the same system at a different time.

Ageing: As highlighted above, all safety-related data has a lifetime and this needs to be explicitly managed. This can involve, for example, purging, deleting and alerting. Ageing can occur as a result of changes external to the system (for example, records of the positions of other aircraft, newly discovered drug interactions, or new software patches), or it can result from internal changes (e.g., valves gradually becoming less responsive, [configuration data](#) becoming out of date, or data schemas evolving over time).

Transformation: Data is often filtered, mapped or aggregated as it moves through systems, sometimes creating new [datasets](#) as a result. [Data properties](#) are not necessarily preserved by these processes. Sometimes data is filtered too much or only some of the data is selected (either deliberately or inadvertently through accident or unintended bias), such as by the selection of only the test runs that succeeded. The main issues are loss of heritage / history / source [information](#). Data can also appear to become something else. Without careful management, the [integrity](#) may become lowered to the lowest common denominator and this needs to be recognized. Additional checks (e.g., [validation](#) checks, credibility checks) or assurance measures may be needed to ensure that required [integrity](#) / assurance is maintained.

Archiving and retrieval: Safety-related data needs to be available when required. Data [accessibility](#) needs to be considered over the complete system lifetime. It is also important to consider what properties of the data need to be preserved and how this affects the choice of storage medium.

Biasing: This is a systemic inaccuracy in data due to the characteristics of the process employed in the creation, collection, manipulation, presentation, and interpretation of data. It is usually an unintentional distortion in the [dataset](#). One example of this is the confirmation bias that may be applied to safety claims, and another example is that synthetic autonomous vehicle training [databases](#) can have issues with artificial data if it is not realistic. Although there is no perfect way of checking for this within the system, [completeness](#), statistical and validity checks on [datasets](#) may help.

Falsification / misinformation: This issue arises where data is created, modified, or deleted either accidentally or deliberately to mislead or misinform potential consumers of that data. Examples from policing and criminal justice might include notes being taken with fabricated times or dates or a crime from the wrong individual in a [database](#) being accidentally added or removed. Another example might be a supplier falsifying quality records for materials or goods. There have been many cases of misinformation related to the Covid-19 pandemic (for example on social media) where people have been deliberately misled, and sometimes this has led to harm (for example, bleach being drunk

as a cure). Possible [mitigations](#) include digitally signing transmitted data, strong access controls, independent fact-checking, and audit records.

Defaulting: Many systems use default or initial values for [data items](#); sometimes in [datasets](#) and sometimes embedded in software. Often these default values are designed to be neutral (e.g., “0”) or unrealistic (e.g., “VOID”). There are essentially two cases:

1. initialisation data which may persist and be mistakenly taken as a real value when in fact it should have been changed; and
2. data that is used when no meaningful value has been assigned (e.g., during data migration or data exchange between systems).

These issues can often be managed through good design of data structures, for example by the inclusion of a validity flag.

Sentinels: A sentinel value is a data value that is used to indicate a special action needs to be taken, typically indicating the end of a record or a [dataset](#). The sentinel value should be one that is not allowable in the [dataset](#) itself but often is not properly considered and may use common sequences (e.g., five zeroes). Sentinels can cause problems in two ways:

1. where they are not recognized and so, for example, processing includes or continues past the sentinel; and
2. where the data itself somehow contains the sentinel value and so processing is erroneously interrupted.

Sentinels can be a particular risk in long-lived systems and [datasets](#). As with defaulting, the management or elimination of this issue may often be achieved through improved data structures.

Aliasing: This is an effect that causes different data to become indistinguishable when accessed; that is, there is only one record when there should be several – for example, two patients with similar names inadvertently sharing a single set of medical records. This could be due to the way the data is filtered, sampled, indexed, stored, or retrieved. The data issues are typically related to loss of resolution leading to similar data points appearing to be identical. Methods to maintain resolution, including use of unique indexes, may be beneficial.

One specific case of aliasing concerns homophones – words which sound the same but are different such as “flower” / “flour” and “bare” / “bear” [1]. If these words are read out over a phone there may be confusion as to which is intended. Location services based on a number of words such as W3W don’t avoid all homophones, and so there may be issues when the words are given verbally in an emergency. The issue is discussed further in Section [section 5.5](#).

Disassociation: This effect is, in some senses, the opposite of aliasing: there are several records when there should only be one. This could occur, for example, if two records are created for the same individual using slightly different names. It could also arise if different systems use different indexing methods and the association between the indexes becomes corrupted. Again, methods to maintain data resolution can be beneficial.

Masking: This issue can arise if a notable proportion of a [dataset](#) is of a poor [data quality](#), for example, if sensors producing the data are faulty or measurements are taken from the wrong source. This poor quality data can mask errors in the way that the system handles the good quality data. One way of protecting against this issue is the generation and use of test sets of appropriate size and quality, although for some applications this may be a non-trivial task.

Incompleteness: Not all the data that is needed is always available; there may be known, and sometimes, unknown gaps or missing data points. The missing data points (“dark data” – see Section 2:??) can be critical and in some cases, more important than the data that is available. Incomplete data can arise, for example, from limitations on how much data can be physically captured (e.g. sampling frequencies, storage / time constraints) or from the unintentional or deliberate darkening of data (e.g. for privacy, security, political or commercial reasons).

Volume: Data can be so large and unstructured that it is not manageable in practicable timeframes. For example, video records of rail track could take days to inspect manually.

Interpretation: Data can be misinterpreted – too much or too little deduced from available data, or data extrapolated incorrectly to derive unsound results. An example is ML data, especially real or recorded data that may not contain critical edge / corner cases.

Distribution: Data can be decentralised, decomposed or distributed across many sources (e.g. channels, databases, websites) and needs to be consistently integrated to make a coherent picture. In the health sector many IT systems often have to work together feeding in different parts of a patient medical record to make a complete health picture. If parts of this distributed data are missing (for instance diagnostic test results) then it is difficult if not impossible to obtain the complete picture, and mistakes may be made. There are several parts to this problem:

Integration – multiple elements of data have to be brought together in a coherent way, addressing aspects such as which data should supercede or replace other data;

Communication – having correct and current information about what data is available and its location so it can be requested; and

Contingency: – what to do if part of the distributed data is unavailable or late.

Further examples of how data may cause issues in many scenarios is given in the book *Data-Centric Safety: Challenges, Approaches, and Incident Investigation* [2].

2.3 Data Properties

???Originally said move to Volume 1, but I think it fits better here. Data properties are used to establish what aspects of the data (e.g., timeliness, accuracy) need to be guaranteed in order that the system operates in a safe manner.

James Inge’s work [3] produced a useful taxonomy of data categories, and went on to look at faults in data. He concluded that a rigid taxonomy of data categories was unhelpful due to various properties or characteristics of the data which vary independently. In short, it is the combination of data category with the required data properties that facilitates safety analysis.

Data categories were discussed Chapter 2:13. Table 2.1 documents a non-exhaustive collection of data properties. Typically, it is the loss of one of these data properties that presents a hazard. This notion of “loss” is dependent on the intended use; for example, what is “timely” for one use may not be for another.

The “Goldilocks” property¹ addresses appropriate sizing and quantity of data. A number of issues have been found to arise when there is too much or too little data. While it is particularly relevant to communications

¹ The authors are aware that some of the names of categories are very culturally dependent. For example, we are informed that the folk-tale of Goldilocks is not well known in Chinese culture but the idiom guòyóubùjǐ, interpreted as “doing too much is just as bad as not doing enough” is well known. The intention of the names is simply to evoke the associations, so users of this guidance are encouraged to use names for the categories that are more appropriate to their cultures.

links, it may have relevance to other areas such as [databases](#) and when people are involved in reviewing or checking data. The property is named “[Goldilocks](#)” as it refers to the need to have not too much, not too little, but just the right amount of data??? LaTeX crashes if I have this footnote on the same page as a longtable. A system where the property was lost involved a high-speed data bus that connected several safety-critical systems. A transceiver of that bus failed and transmitted random noise. The receivers employed parity checks and cyclic redundancy check, but the system had been designed to eliminate occasional [data errors](#). When random noise filled the bus, several apparently valid messages were created every second, resulting in potentially lethal behaviour.

In a [hazard and operability study \(HAZOP\)](#) carried out during 2020, based upon the [HAZOP](#) guidewords in this guidance, the facilitator realised that certain failure modes had not been identified by the [HAZOP](#) team. In addition to the issue of system overload already discussed, those omissions also concerned system behaviour following data rejection. In these cases, bad data was detected and rejected, but the consequences of data rejection over an extended period had not been considered.

The [Goldilocks](#) property is related to the data volume problem discussed in [section 2.2](#), however, given the importance of data sizing and the experience of real-world incidents this is now a separate property.

2.3.1 The analysability property

The [analysability](#) property recognizes that data is now highly complex and extensive, often large scale, and distributed. And yet, for safety purposes we need to make sure it is of suitable quality and able to support system goals. We therefore need to ensure that it is possible to analyse it for key characteristics and establish meaningful results using tools or other means. This property is related to explainability and may be performed by the same set of tools or techniques.

2.3.2 The explainability property

[Explainability](#) describes the ability to establish what the purpose and effect data (and especially changes to data) has on a system and explain this to relevant [stakeholders](#) in terms that they can understand. It is particularly important for learning and AI-based systems. This could be, for example, [ML](#) training data or system [configuration data](#). This property is related to [analysability](#) and may be performed by the same set of tools or techniques.

Table 2.1: Properties of data

Property	Abbreviation	Description
Integrity	I	The data is correct, true and unaltered
Completeness	C	The data has nothing missing or lost
Consistency	N	The data adheres to a common world view (e.g., units)
Continuity	Y	The data is continuous and regular without gaps or breaks
Format	O	The data is represented in a way which is readable by those that need to use it
Accuracy	A	The data has sufficient detail for its intended use
Resolution	R	The smallest difference between two adjacent values that can be represented in a data storage, display or transfer system

Continued on next page

Table 2.1: Properties of data (continued)

Property	Abbreviation	Description
Traceability	T	The data can be linked back to its source or derivation
Timeliness	M	The data is as up to date as required
Verifiability	V	The data can be checked and its properties demonstrated to be correct
Availability	L	The data is accessible and usable when an authorized entity demands access
Fidelity / representation	F	How well the data maps to the real-world entity it is trying to model
Priority	P	The data is presented / transmitted / made available in the order required
Sequencing	Q	The data is preserved in the order required
Intended destination / usage	U	The data is only sent to those that should have access to it
Accessibility	B	The data is visible only to those that should see it
Suppression	S	The data is intended never to be used again
History	H	The data has an audit trail of changes
Lifetime	E	When does the safety-related data expire
Disposability / deletability	D	The data can be permanently removed when required
Goldilocks	G	The data is just the right size – not too much and not too little
Analysability	Z	The data (including any metadata) is of a suitable size, type and format to enable it be usefully analysed
Explainability	X	The data can be meaningfully explained, by a suitable mechanism, to those who need to understand it
Clarity	K	The data is visible, unmasked, not hidden or obscured and not camouflaged

Table 2.2 illustrates where the data issues discussed in [Section 2.2](#) can result in loss of one or more data properties (x = potential loss of property).

Table 2.2: Properties that can be lost through issues

	Integrity	Completeness	Consistency	Continuity	Format	Accuracy	Resolution	Traceability	Timeliness	Verifiability	Availability	Fidelity / representation	Priority	Sequencing	Intended destination / usage	Accessibility	Suppression	History	Lifetime	Disposability / deletability	Goldilocks	Analysability	Explainability	Clarity
Issue	I	C	N	Y	O	A	R	T	M	V	L	F	P	Q	U	B	S	H	E	D	G	Z	X	K
Fluidity								x		x		x						x			x	x	x	x

Continued on next page

Table 2.2: Properties that can be lost through issues (continued)

	Integrity	Completeness	Consistency	Continuity	Format	Accuracy	Resolution	Traceability	Timeliness	Verifiability	Availability	Fidelity / representation	Priority	Sequencing	Intended destination / usage	Accessibility	Suppression	History	Lifetime	Disposability / deletability	Goldlocks	Analysability	Explainability	Clarity
Issue	I	C	N	Y	O	A	R	T	M	V	L	F	P	Q	U	B	S	H	E	D	G	Z	X	K
Reuse		x				x		x	x	x		x			x		x	x	x	x	x		x	
Ageing		x				x			x			x			x	x	x	x	x	x			x	
Transformation	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Ownership								x		x	x				x	x		x					x	
Archiving / retrieval															x	x	x	x	x	x		x		x
Biasing	x	x	x	x		x			x	x		x	x	x								x	x	
Falsification / misinformation	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Defaulting	x	x	x	x		x						x										x		x
Sentinels	x	x		x		x								x								x		x
Aliasing	x	x		x		x	x	x		x	x	x										x	x	x
Disassociation	x			x		x		x	x	x	x	x	x	x	x	x	x	x	x	x		x	x	x
Masking	x	x	x	x		x				x		x										x	x	x
Incompleteness	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Volume										x											x	x	x	x
Interpretation	x	x	x		x	x	x		x	x		x										x	x	
Distribution	x	x	x	x				x			x		x	x	x	x		x		x		x	x	x

2.4 Data in System Lifecycles

Like other components of a safety-related system, the safety dependency of data is dictated by the context in which it is used and the causal links that become established where loss of one or more of the required properties can contribute to hazardous system states. For example, a given [dataset](#) (say [configuration data](#)) could be used in a number of separate contexts such as:

- prototyping a system to demonstrate solution feasibility of a safety-related system;
- development testing of a safety-related system; and
- live operational use of a safety-related system.

In these cases, the [dataset](#) is the same but the context of its use changes the safety significance and therefore the level of assurance that it may require. It follows that the [DSAL](#) of a [dataset](#) is also predicated

on where and when in the lifecycle the [dataset](#) will be applied.

To illustrate this concept, a number of generic model lifecycles are discussed below. Note that these are not intended to be prescriptive or mandate the use of any particular model. Instead, they are being used to illustrate how the Data Safety Management Plan could articulate these lifecycle considerations.

Development: the diagram in [Figure 2.2](#) represents a typical development lifecycle using an iterative development approach². In this model there are key phases as the system transitions from concept through to testable executable code. The process is iterative in that several cycles of functional elaboration, design, development and test may be run and these typically will focus on the areas of the system that bear most technical risk or comprise the key functional use cases so the client gets early visibility of the system. This early awareness allows feedback to be provided into the next iteration to help steer the solution to the client's actual needs. Traditional waterfall implementation can map onto this model on the basis that there is only one iteration in each phase and all activities in one phase need to be completed before progressing to the next.

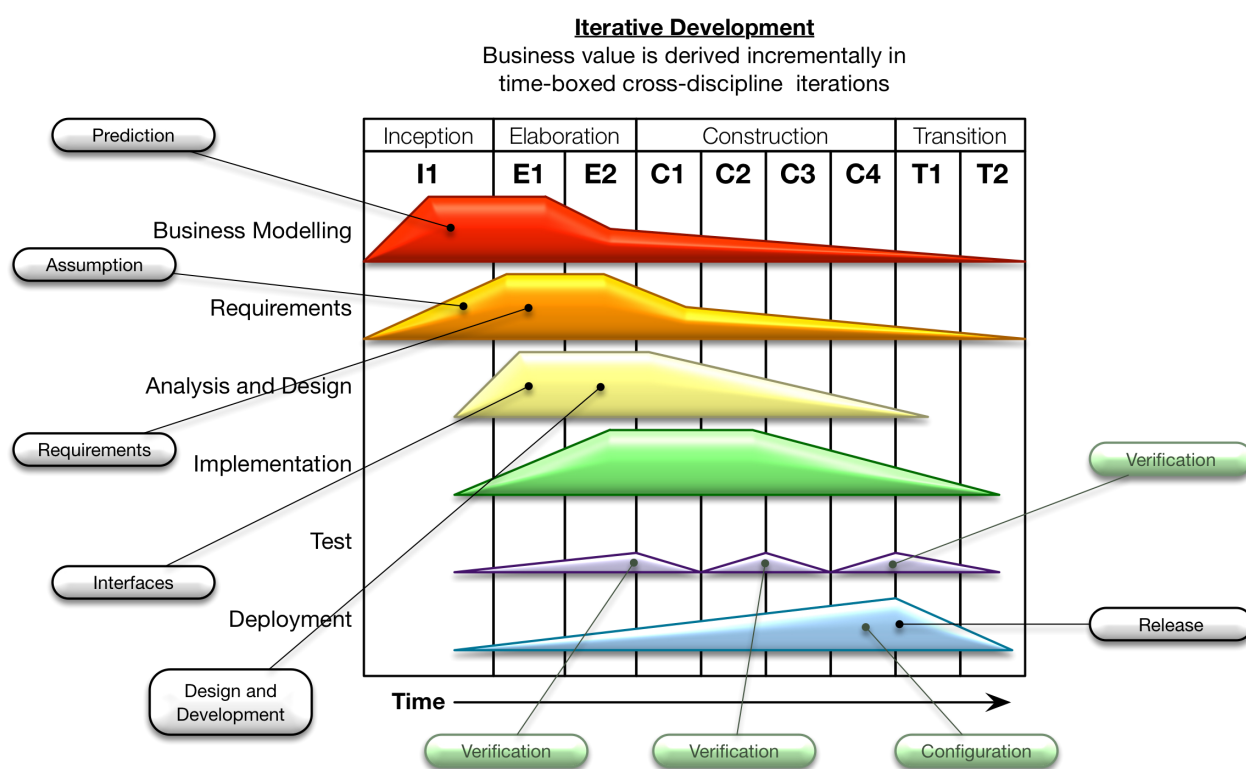


Figure 2.2: Development lifecycle

The model itself may vary depending on the specific needs of the project but the diagram illustrates that different data categories become significant at different points of the process.

Operational Once a system has been developed it will move into an operational lifecycle or indeed, if data safety has not previously been considered for an enterprise, then the system could already be in operational use. These operational lifecycles tend to be cyclical in nature; the diagram in [Figure 2.3](#)³ illustrates a typical model.

Again, specific data will come into play at different periods in the process. Documenting the relationship

² The diagram is based on [International Business Machines Corporation \(IBM\)](#)'s Rational Unified Process, an iterative software development process framework. The original diagram is in the public domain.

³ ITIL is a registered Trade Mark of AXELOS Limited. All rights reserved.

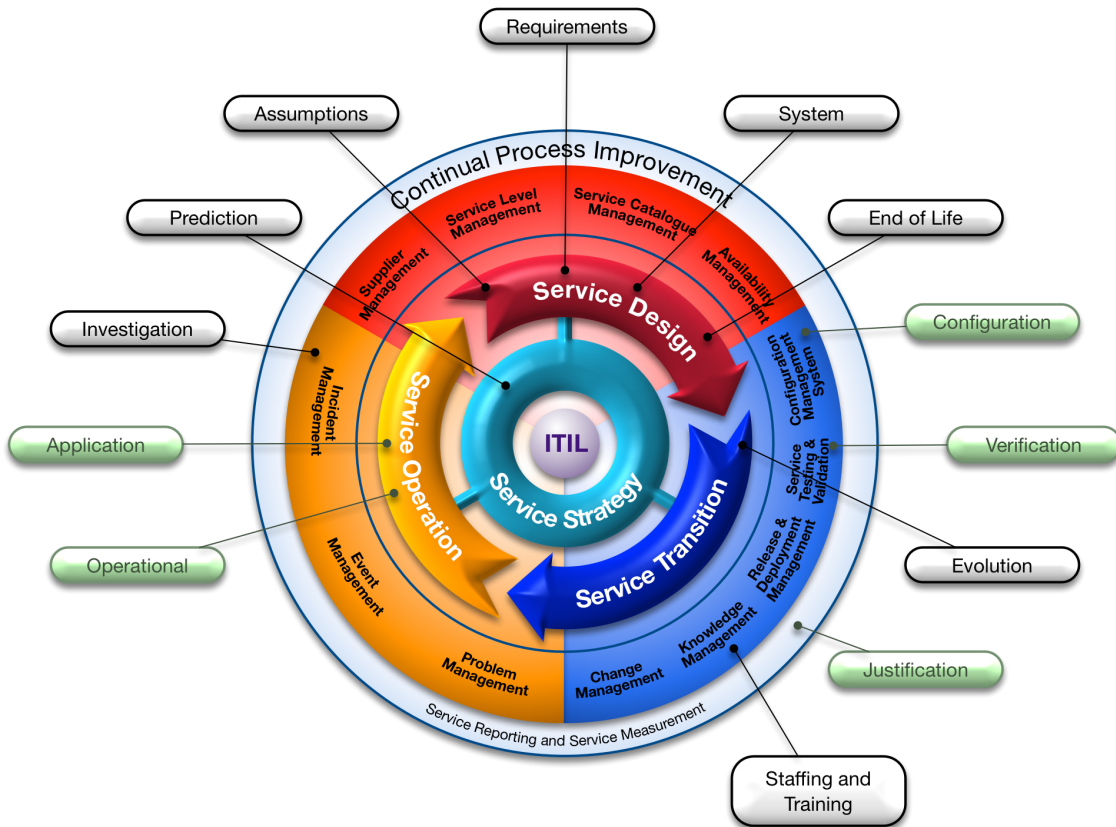


Figure 2.3: Operational lifecycle

between process steps and data categories will therefore give clarity as to when a particular assurance technique needs to be applied.

Data supply chains The previous models relate to typical system supply and operate perspectives but there are also other data supply chains where a number of organizations engage in the procurement and use of safety-related data. These processes may include the development and operational lifecycles but a different model is required to fully represent the wider processes that are being employed. The diagram in [Figure 2.4](#) shows such a model representing a data acquisition lifecycle.

This model represents the interactions between three key organizations:

- The commissioning user: the organization that has the need for the data;
- The data provisioner: the organization that will fulfil that need for data; and
- The data acquirer: the organization employed by the Data Provisioner to carry out physical collection of data.

Note that these may be three separate organizations, or they may be separate business units within the same, larger, organization.

In this supply chain, the commissioning user is a consumer of the data and the data acquirer is a producer of data. The data provisioner acts both as a consumer (from the data acquirer) and producer (to the commissioning user) of data. Similarly, an organization that augments [datasets](#) is both a consumer and producer of data in the supply chain.

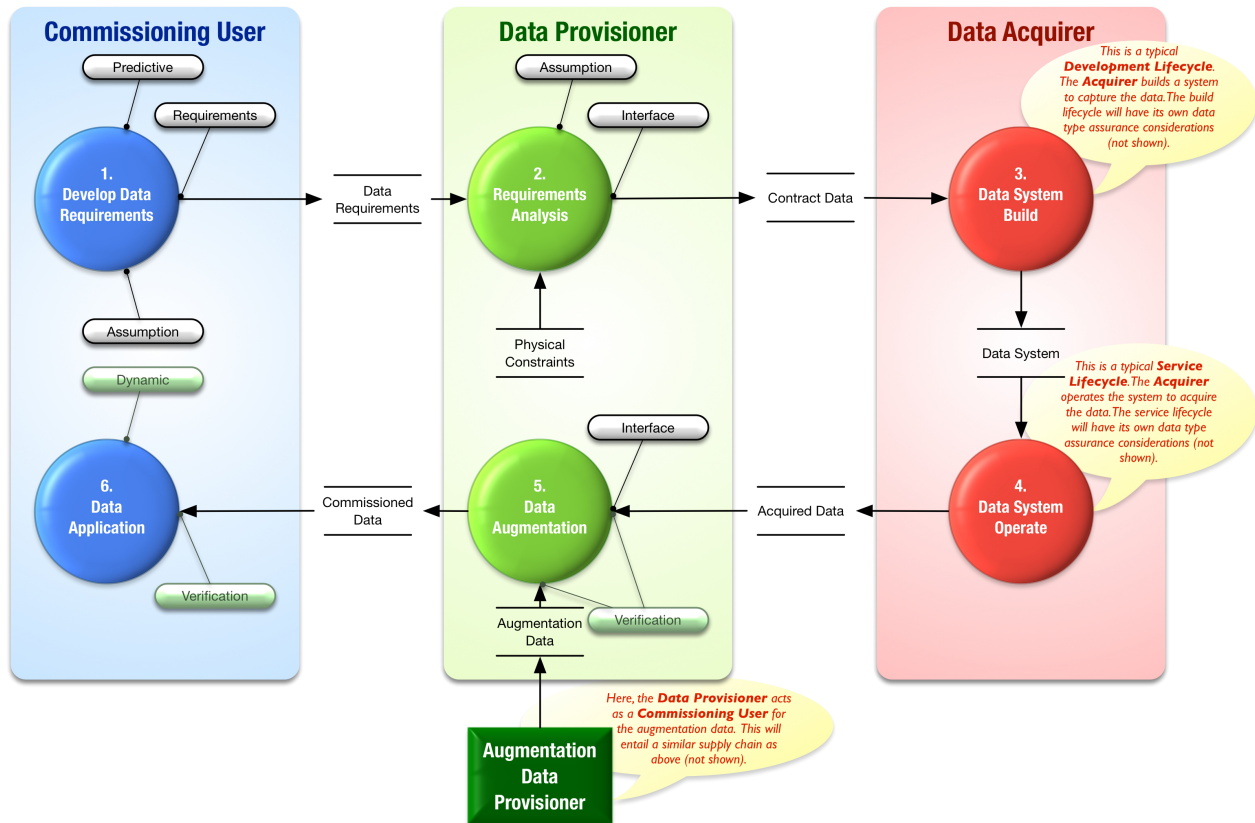


Figure 2.4: Data supply chain

The commissioning user requirement analysis is the key process step where the commissioning user's expectations for data are agreed with the data provisioner. The requirements may be adjusted because of physical constraints (e.g., loss of precision because of physical measuring device constraints) and may include additional requirements to augment the captured data with additional **information** (e.g., airport codes added to a measurement of a given runway length).

The data provisioner may employ a data acquirer to capture the data (e.g., to carry out a physical survey of a site). The acquisition phase may itself require a specialised system to be built to perform the capture and data refinement to meet the data provisioner's specifications. Such systems will then themselves be subject to the development lifecycle model considerations discussed above. Likewise, the data augmentation phase may require further system development processes or indeed, could trigger an instance of the model again as the data provisioner acting as a Commissioning User.

Acquired and augmented data is then fed into the operational system that has been built for providing the service of generating the commissioned data. This system in its service provision role would then typically follow the operational lifecycle process model discussed earlier.

2.4.1 Tool Assurance

Tools in this context are considered anything that automates all or part of a process, for example, data creation or data transformation. Test tools are also included (i.e., the term is not limited to parts of an operational system).

Tools can impact data safety in different ways, depending on both their function and how they are to be

used. For tools to be considered fit for purpose it is necessary to show that the tool meets its requirements in the context in which it is to be used. The activity to ensure a tool is fit for purpose is usually called “tool qualification”.

The first step is to define the purpose for which the tool is required to be fit. Once that is done, and the tool's requirements are specified, there are three main strategies available for qualification:

- Use evidence of a previous certification of the tool by a trusted third party (unlikely to be available in most industry sectors);
- Base tool qualification on the practices used when designing and developing the tool (only practicable for tools developed within the organization); and
- Use one of the available industry-specific guidance documents that admit [commercial off-the-shelf \(COTS\)](#) solutions, e.g., EUROCAE Document ED-215 (RTCA/DO-330) [4].

Further details on tool qualification are presented in Part 2 Chapter 10.

2.4.2 Test Data

The generation of suitable test data is critical to verification of a safety system. The test data must include both representative “normal” values based on intended usage and also values which push at, and beyond, normal use to provoke [hazards](#) that the system might produce. This latter type of test data is particularly hard to generate; generally it must be credible, yet it must stress the system to react in a way that the preservation of safety properties can be assessed.

In general, all the properties of the test data should be considered and an assessment made as to whether breaking a property (e.g., introducing corrupt or late data) would cause a problem to the system. If it does, then specific test data should be produced to facilitate testing of this potential problem.

Some suggestions for test data for safety-related systems are:

- Use of values on or around boundaries;
- Use of extreme values, way beyond what could be reasonably expected;
- Use of typical “everyday” values / sets;
- Some realistic but unexpected values;
- Try combinations of data values or [data items](#) that are problematic together (e.g., inconsistent);
- If possible, use some values known to have caused problems in the past;
- Where appropriate, use values related to timing, rollover or date boundaries;
- Where possible, use white box values (i.e., those derived from an understanding of the system);
- Use a set of values with drift or bias over time;
- Use [datasets](#) with particular statistical properties (e.g., distribution, patterns etc.);
- Use data which has incorrect formatting, ordering, or out of sequence, etc.; and
- Try data which has repeated sets of values or pseudo-random characteristics.

Typically very complex test data is derived from recorded live feeds of real data flows. While this data can be extremely useful for regression purposes, it should be recognised that it is unlikely to contain many outlying or boundary [data items](#). Therefore it may need to be modified to test any hazardous situations; this modification can be difficult and may require sophisticated tools to both ensure correct properties and injection of the intended faults (for instance to introduce a statistical bias to the data).

Simulator / emulator derived values can be useful, but again the issue is how realistic the values are: often the [accuracy](#), [resolution](#) or timing of simulated values may be different to real data.

Coverage with test data is something to consider. Sometimes the same [dataset](#) is used for multiple test scenarios, when in fact it is not stressing all of them to the same degree. Test data coverage can be collected over requirements, code or design, but it is important not to forget hazards: coverage of the hazards and mitigations identified in the [hazard log](#) is a key aim.

In general some measure of the quality and suitability of the test data can be useful. This could be based on statistical properties, coverage of hazards or coverage of requirements.

Test data must show continued relevance, through systems evolution and over time. It is good practice to build up extensive regression suites containing coverage of all detected problems to date.

2.4.3 Interfaces with Existing Assessments

2.4.3.1 Data and Software

Although most people feel they have an intuitive understanding of the difference between software and data, upon closer examination the boundary is not always as clear as it may first appear.

Consider, for example, Java bytecode, which is operated on by a Java virtual machine. From one perspective, it could be argued that the Java bytecode is simply data. By extension, it could also be argued that the Java source code is also just data. This type of argument can be extended to suggest that any software can, at least from one viewpoint, be considered as data. Conversely, think about the data used in a 3D printer, perhaps to produce a part for an aircraft. This data could be viewed as a program for the printer; that is, it could potentially be viewed as software. This type of argument can easily be extended across a range of situations, especially those relating to [configuration data](#).

While they are interesting, and potentially important, these philosophical considerations should not detract from the practical issue: there are some aspects of data (using the term in a generic sense) that are often not explicitly addressed in standards. These are a consequence of features that are more readily apparent in data than in software. Examples include:

- It is easy for data to be reused in a range of contexts and despite appearances it is not trivial to translate an assurance argument that the data is fit for purpose from one context to another.
- It is not always clear who owns or is responsible for data, especially when data is shared and processed amongst a collection of disparate systems.
- Data often has a [lifetime](#), that is a time after which it is no longer valid. This may be a strict cut off, or a more gradual degradation in the utility (or applicability) of the data.
- There is often a default value for data. While this can make systems easier to use and hence more productive it can be difficult to identify a default value that is appropriate for all circumstances.

- It can be easy to change data. In some circumstances this can give rise to a temptation to make uncontrolled and potentially untested changes. It can also allow data to be fraudulently changed after an accident.

In summary, data and software are closely related and, as such, need to be considered together in system engineering activities, including system safety analyses. However, data and software emphasise different facets of risk and they are susceptible to different mitigation approaches; this means there is also a need to adopt a data-focused perspective. It also means that [software assurance levels](#) cannot be mapped directly to [DSALs](#).

2.4.3.2 Data Safety and Security

When generating high-level processes and techniques to manage the risks posed by data, it is worthwhile understanding the difference between the safety risks posed by accidental failure to preserve Data Properties and the security risks posed by actors maliciously undermining the properties of data.

The relationship between safety and security, as engineering concepts, can be summarised by their relationships to cultural, developmental and aspirational properties of systems development.

Culturally, embedding both safety and security into an organization is seen as a key strategic goal for creating systems that are both safe and secure. Developmentally, safety and security are quality factors, generating transverse requirements that impact the entire system. Most importantly, at the aspirational level, both safety and security have the common goal of preventing harm from accidental and malicious interventions respectively.

For an organization aiming to create systems that are both safe and secure, these connections can be both a benefit and a burden. The shared goal of preventing harm means that both quality factors seek to identify routes to harm through analysis of the system being developed. This can result in shared processes and tools, which in turn can save time and money during systems development. However, safety and security interact in a more volatile way at the functional level. Security failings can undermine the safety case for a system and, conversely, safety requirements can prevent the implementation of standard security solutions. For example, the German government published a report in 2014 into a fire at a steel works caused by a cyber attack that resulted in the control system being placed into an unsafe state and the safety system being unable to intervene (Section 3.3.1 of [5] - in German). In addition, “fail-safe” states can often leave a system with exposed security vulnerabilities.

These links between safety and security infer that there are connections between the sub-categories of data safety and [information](#) security: both attempt to take a data-centric view of the system of interest in order to improve the associated quality factor; and both attempt to prevent harm through the preservation of the properties of data within that system.

In the security domain, the three key properties of data considered are [confidentiality](#), [integrity](#), and [availability](#). [Confidentiality](#), the failure of which is termed “[information](#) disclosure” in the Microsoft Security Model, [6] is typically not a safety concern as, without malicious intent, [information](#) sharing is not inherently unsafe. However, when considering systems where [confidentiality](#) is an important property, the interaction between data safety and security cannot be trivially resolved. For example, accidental disclosure of [information](#) can form part of a causal chain which leads to harm from a malicious actor.

Data [integrity](#) is a critical property for both domains. The Microsoft Security Model describes malicious removal of the property of [integrity](#) as “tampering”. Whether by accident or through malicious intent, the

potential harm from loss of data [integrity](#) can be disastrous to a safety-critical system, from the values of drug dosages to control system parameters.

Data [availability](#) is also important to both domains. Loss of [availability](#), or “denial of service” in the Microsoft Security Model, is another property that can be lost accidentally or through malicious intervention. Loss of [availability](#) prevents systems from functioning properly and can result in undefined behaviour if not mitigated by design.

Further guidance on the integration of safety and security can be found in a code of practice published by the IET [7]. The Code of Practice is written for engineers and engineering management to support their understanding of the issues involved in ensuring that the safety responsibilities of an organization are addressed, in the presence of a threat of cyber attack.

3 The Data Safety Tool: RADISH (Discursive)

Technology is nothing. What's important is that you have a faith in people, that they're basically good and smart, and if you give them tools, they'll do wonderful things with them.

Steve Jobs

3.1 An introduction to the tool

RADISH (The Risk Assessor for Data Integrity and Safety Hazards) is a software tool being developed by [Mission Critical Applications Limited \(mca-ltd.com\)](https://mca-ltd.com), to assist a data safety practitioner developing a data safety case using the guidance, by:

- recording the decisions that are made;
- automating parts of the [data safety assessment](#) process; and
- helping the practitioner to choose between the risk mitigations that are recommended by the guidance, given the nature of each risk.

For more information, and to access to the RADISH tool, visit data-safety.tech/tooling.

As illustrated in [Figure 3.1](#), the RADISH tool is a central repository of information about the data safety case for a project. RADISH guides data safety engineers to identify the data artefacts in the system, and the safety properties that are important for each artefact. It then calculates the highly recommended and recommended mitigation techniques following the guidance, allowing the engineer to choose those which should be in the system requirements those which should not, and record the justifications for those choices.

RADISH can generate reports showing the risks that have mitigations, and those where more work is needed, giving project management a view of the state of the data safety process. RADISH can also generate a report that can be included in a data safety case to support the system safety argument, and this can continue to evolve throughout the project lifecycle of development, deployment and maintenance.

3.2 A caveat

The tool described in this appendix is usable, but is not yet at production standard. The developers are working towards a release in 2026, and are looking for programmes to use as test vehicles. To have a programme considered as a possible test vehicle for the tool, please contact the developers at [Mission Critical Applications Limited \(radish@mca-ltd.com\)](mailto:radish@mca-ltd.com).

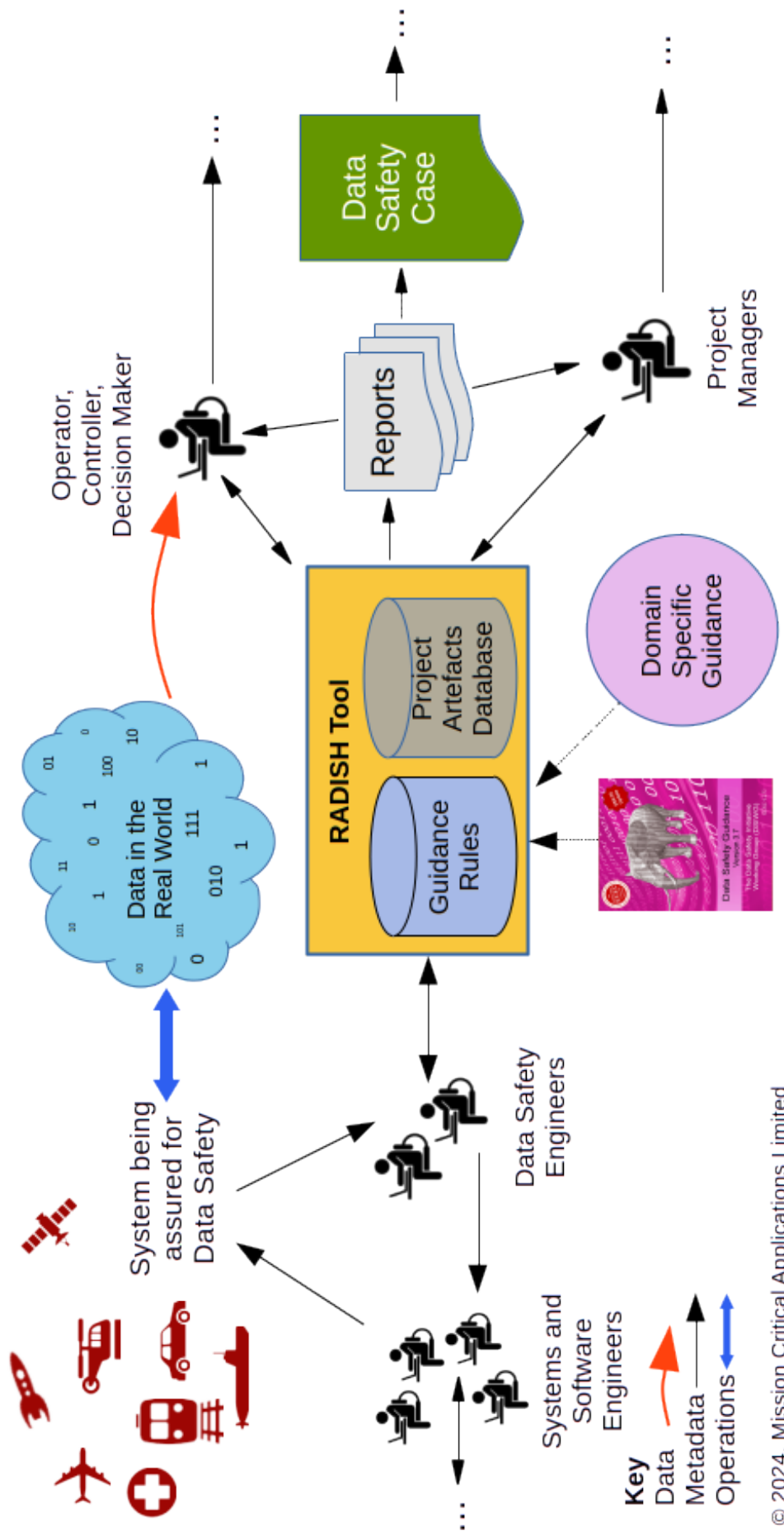


Figure 3.1: RADISH within the project environment

4 Covid-19

The public health community wants a safe and effective [COVID-19] vaccine as much as anybody could want it. But the data have to be clear and compelling.

Michael Osterholm

4.1 Covid-19 and Data

The Covid 19 crisis has highlighted a number of areas where better data and data management could have improved outcomes and hence reduced the death toll. Such pandemic-related data is therefore very much safety-related data.

Some issues related to Covid-19 data are:

1. The lack of [consistency](#) and standardisation in handling the data, e.g:
 - a. In predictive models where many assumptions may be wrong, or the algorithms inappropriate
 - b. Presentation of statistics in a selective or misleading way
 - c. Methods of data collection which may be selective or incomplete. For instance reporting on the number of positive test cases is always misleading, as many people may be asymptomatic with the virus so never get tested.
 - d. Calculations and filtering
 - e. Allowances for delays in collection or processing
 - f. Intentional and unintentional bias
 - g. Use of averaging (e.g. moving averages) and smoothing of plots hiding sudden increases
 - h. Loss of data (e.g. in the UK Test and Trace system due to old versions of Excel)

All of which have prevented any meaningful comparisons internationally, even between countries in Western Europe. They also lead to public confusion; this in turn leads to mistrust and a refusal to abide by guidance and regulations.

2. The poor data within the Test and Trace systems is a major factor in the failure of these systems. If data is not accurate, timely or complete then contacts cannot be traced in time, and the effort put into the activity is wasted. These systems in the UK need huge improvement as there is currently low contact performance and hence very poor outcomes [8].

4.2 Systems Involved with Covid-19 Data

Many systems have been created or re-deployed to help manage the pandemic. These systems consume and produce vast amounts of data, some of which is critical and could affect safety of individuals or the general population. Some identified systems are shown in [Table 4.1](#). For each of these, it is worth thinking through some basic data failure modes, e.g. data is lost, late, incorrect or incomplete. For instance if we are running an infection / spread model and we feed it with stale data, then its predictions will clearly be inaccurate.

Table 4.1: Systems involving data used to manage the pandemic

Analysis of air flow and particles	Satellite imagery (Wuhan)	Video conferencing	Risk assessment systems
Infection / spread models (inc new variants)	Itinerary systems	Remote consultation systems	Computational bioinformatics tools
Infra-red / thermal cameras	Infection testing systems	Ventilators / other patient management devices	Appointment systems
Track and Trace apps	Antibody testing systems	Personal risk profiling apps / systems	Border Control / Quarantine systems
Track and Trace back office systems	Drug trials systems / data	Allocation / reservation / booking systems	Risk profiling / prioritising for vaccination
Track and Trace service	Ventilation models & UV sterilisers	Models of built environments	Digital Twins (systems and biological: lungs, etc.)
Supply chain systems	Behavioural models	Safety analyses (STAMP/STPA), etc.	(Automatic) cleaning systems
Virus aerosols modelling	Analysis of delays in system of reporting / actions	Modelling / public perception of the disease	Virus shedding models
Vaccination booking / tracking / monitoring	Sanitiser systems	Lockdown easing models	Vaccination Passports
Vaccination production data	Vaccination trials and reporting data	Vaccination “Yellow Card”	Cross-system data sharing
PPE testing results	Data used to inform public perceptions	No coordination across international boundaries – incompatible systems	Use of blockchain to validate Covid and vaccination status

4.3 Falsification / Misinformation of Data

One serious and perhaps unexpected aspect of the pandemic is that of misinformation. There are people either in denial of the virus’s dangers, refusing to socially distance or refusing vaccinations. Reasons for these behaviours have generally been driven by intentional misinformation, ignorance, superstition, or economics. Marianna Spring, the BBC’s specialist reporter covering disinformation and social media put the problem of misinformation succinctly when she stated “The problem with misinformation is that it is popular.” [9]. Methods need to be devised or improved to prevent this effect, and to restore trust in carefully managed data.

4.4 Rumsfeld’s known unknown and unknown unknown data conundrum

It is clear that until China reported to the world that Covid 19 had emerged as a threat, that data about the virus was an unknown unknown. However, we know there are thousands of viruses in animals that could pose a threat. These all need analysis and it may be possible to use massive computer analysis of genetic data to identify likely new threats.

detail on this topic *"Data Safety in Virus Outbreaks – Lessons learnt and Recommendations"* [10].

4.5 Learning

It is important to learn from these deficiencies because, while Covid 19 has brought tragedies with it, it is unlikely to be the last, or indeed, the most dangerous virus we will face. We must do better next time.

This page is intentionally blank

5 Incidents and Accidents (Discursive)

Hiding within those mounds of data is knowledge that could change the life of a patient, or change the world.

Atul Butte

5.1 General

The following ‘war stories’ describe incidents and accidents in which data is considered to have been a contributory factor. A data perspective has been taken to demonstrate the need for data to be given equal footing alongside software, hardware and human factors. The items described here have been arbitrarily selected; the collection is not intended to be exhaustive.

Note: The analysis presented here has no legal standing whatsoever. The purpose of this section is not to discredit, contradict or undermine any existing accident or incident analysis; the aim is simply to view these incidents and accidents from a data perspective. Where possible accident reports have been referenced with the role of data highlighted. All references have been taken at face value and not independently verified.

Two tables are presented to assist the reader in finding incidents and accidents of interest:

- [Table 5.1](#) lists the issues and accidents according to the domain where they occurred.
- [Table 5.2](#) lists them in reverse chronological order of incorporation – those entries that have been added to this document most recently appear at the top of the table. This table can also be used to find examples relevant to specific [data properties](#).

Table 5.1: Incidents and accidents (by domain)

Ref.	Title	Ref.	Title
<i>Accountancy</i>			
5.2	Post Office Horizon System		
<i>Air</i>			
5.9	Boeing 737 MAX 8 crashes	5.30	LOT Flight 282
5.17	Turkish Airlines A330	5.32	Comair Flight 5191
5.19	Qantas Boeing 737 Take-Off	5.33	Überlingen Mid-Air Collision
5.20	Qantas Boeing 737 Loading	5.37	Crash into Nimitz Hill, Guam
5.23	Boeing 737-33A at Chambery Airport, France		
<i>Air (Defence)</i>			
5.15	A400M Torque Calibration Parameters	5.24	Loss of Hermes 450
5.22	Loss of MQ-9 reaper		
<i>Air (Other)</i>			
5.12	Loss of Irish rescue Helicopter		
<i>Defence</i>			
5.3	Battle of Agincourt	5.34	Fort Drum Artillery Incident
<i>Internet</i>			
5.6	Java log4j Library Vulnerability		
<i>Maritime</i>			
5.21	Grounding of <i>Navigator Scorpio</i>	5.29	Grounding of <i>The Pride of Canterbury</i>
5.26	Grounding of <i>Sichem Osprey</i>	5.31	Annabella container ship – Baltic Sea
<i>Maritime (Defence)</i>			
5.16	Royal Navy Submarine		
<i>Medical</i>			
5.5	What3Words systemic ambiguities	5.18	Dallas Hospital Ebola Incident
5.7	Immensa False Negative Covid-19 polymerase chain reaction (PCR) Tests	5.25	Advocate Lutheran Hospital
5.8	Covid-19 Test Results Silently Deleted by Excel	5.28	Cedars-Sinai Medical Center Scanner
<i>Oil & Gas</i>			
5.39	Lake Peigneur Drilling Accident		
<i>Policing</i>			
5.5	What3Words systemic ambiguities	5.35	Early release from Washington State Prison
5.14	Interception of Communications		
<i>Rail</i>			
5.11	Cambrian Line Data Loss	5.38	San Bernardino derailment and pipeline rupture
5.27	Near Collision of Trains, Cootamundra		
<i>Space</i>			
5.4	Gemini V	5.13	Loss of Schiaparelli Mars Lander

Continued on next page

Table 5.1: Incidents and Accidents (continued)

Ref.	Title	Ref.	Title
5.10	Loss of Soyuz-2.1b rocket carrying Meteor-M 2-1 weather satellite	5.36	Mars Climate Orbiter

Table 5.2 provides a summary of the items considered in this appendix. More information on each item is then presented in the following paragraphs.

Table 5.2: Incidents and accidents

Ref.	Title	Summary	Domain	Year	Data Property
5.2	Post Office Horizon System	Non-atomic transactions and other errors in accounting software lead to false prosecutions, lost livelihoods, and suicides.	Accountancy	1999	Integrity, completeness, traceability, verifiability, history
5.3	Battle of Agincourt	French misconceptions about English longbow led to devastating losses	Defence	1415	Integrity, completeness, accuracy, fidelity / representation, timeliness
5.4	Gemini V	Erroneous “schoolchild knowledge” that Earth rotates 360° per day sends spacecraft off course	Space	1965	Accuracy, traceability, fidelity / representation
5.5	What3Words systemic ambiguities	What3Words use of homophones and plurals is alleged to have created difficulties in providing emergency services to persons in distress.	Policing and Medical	2021	Integrity, consistency, aliasing
5.6	log4j Java library vulnerability	Critical zero-day vulnerability affecting Apache Log4j2 java library	Internet	2021	Integrity
5.7	Immensa False Negative Covid-19 PCR Tests	43,000 people with Covid-19 mistakenly given negative PCR results	Medical	2021	Integrity, accuracy, traceability, verifiability
5.8	Covid-19 test results silently deleted by Excel	Importing Covid-19 test results into an Excel file truncated the data after 65536 records	Medical	2020	Integrity, completeness, timeliness, availability, fidelity / representation

Continued on next page

Table 5.2: Incidents and accidents (continued)

Ref.	Title	Summary	Domain	Year	Data Property
5.9	Boeing 737 MAX 8 crashes	On two occasions a single faulty angle of attack (AoA) sensor repeatedly commanded the nose down, leading to the aircraft flying into the sea / ground	Air	2018 & 2019	Integrity, completeness, accuracy, availability, verifiability, fidelity / representation
5.10	Loss of Soyuz-2.1b rocket carrying Meteor-M 2-1 weather satellite	The launch vehicle and satellite were lost because programmers gave coordinates for the wrong cosmodrome	Space	2017	Fidelity / representation, integrity, verifiability
5.11	Cambrian Line Data Loss	Speed restriction data failed to be passed to trains, placing pedestrians on level crossings at risk.	Rail	2017	availability, history
5.12	Loss of Irish rescue Helicopter	At the time of writing, the investigation is continuing; possible controlled flight into terrain; possible issues with terrain / obstacle databases . Four fatalities.	Air	2017	Fidelity / representation, completeness, accuracy
5.13	Loss of Schiaparelli Mars Lander	High rates led to the saturation of the inertial measurement unit (IMU) ; the lander prematurely believed it was on the ground and released its parachute; the lander was lost. The high rates should have been expected, but were not due to modelling deficiencies.	Space	2016	Fidelity / representation, integrity
5.14	Interception of Communications	Incorrect data was disclosed during an investigation into indecent images. A welfare check was delayed on a child believed to be in crisis.	Policing	2015	Integrity
5.15	A400M Torque Calibration Parameters	A software update apparently wiped the engine torque control parameters. Aircraft crash; four fatalities.	Air (Defence)	2015	Completeness
5.16	Royal Navy Submarine, Trawler <i>Karen</i>	A royal Navy submarine snagged the fishing gear of the trawler <i>Karen</i> . The trawler was dragged backwards at about seven knots and suffered structural damage.	Maritime	2015	Resolution, integrity
5.17	Turkish Airlines A330	Inaccurate navigation data, relating to runway location, led to touchdown with left main gear off the paved surface. Aircraft written off.	Air	2015	Accuracy, timeliness, verifiability

Continued on next page

Table 5.2: Incidents and accidents (continued)

Ref.	Title	Summary	Domain	Year	Data Property
5.18	Dallas Hospital Ebola Incident	A man suffering from Ebola was mistakenly sent home from a Dallas hospital. He later returned to hospital, was diagnosed but died; two nurses contracted Ebola but survived.	Medical	2014	Completeness, format
5.19	Qantas Boeing 737 Take-Off	Two independent and inadvertent data entry errors meant weight used when calculating take-off performance was 10 tonnes less than actual weight. Tail strike.	Air	2014	Integrity, verifiability
5.20	Qantas Boeing 737 Loading	Default settings meant that children were incorrectly recorded as adults, resulting in incorrect aircraft weight and balance. Take-off safety speed was exceeded by about 25 knots.	Air	2014	Completeness, fidelity / representation
5.21	Grounding of <i>Navigator Scorpion</i>	The <i>Navigator Scorpion</i> was sailing with out-of-date charts, the planned route was not checked and positional fixes were not taken as often as required. The vessel was grounded, but refloated on the rising tide, with no damage. After the event, false information was added to the navigation chart.	Maritime	2014	timeliness, verifiability
5.22	Loss of MQ-9 reaper	The ground control station (GCS) station was misconfigured following a change from MQ-1 to MQ-9 operations. The misconfiguration was not spotted. It caused any throttle position aft of full forward to command negative thrust. The aircraft decelerated below stall speed and impacted ground in unpopulated area.	Air (Defence)	2012	Consistency, verifiability
5.23	Boeing 737-33A at Chambéry Airport, France	Misuse of Electronic Flight Bag results in tail strike	Air	2012	Timeliness, suppression, lifetime
5.24	Loss of Hermes 450	While attempting an automatic landing the unmanned air system (UAS) self-aborted. This abort was due to an incorrect set-up parameter that had been loaded by the crew. The crew elected to intervene rather than let the UAS self-recover. The air vehicle hit a new, unoccupied hangar; it was ultimately deemed “non-repairable”.	Air (Defence)	2011	Integrity

Continued on next page

Table 5.2: Incidents and accidents (continued)

Ref.	Title	Summary	Domain	Year	Data Property
5.25	Advocate Lutheran Hospital	An infant boy died after a series of medical errors: incorrect information was entered into an electronic intravenous order; automatic alerts had been turned off; and a bag was mislabelled.	Medical	2010	Integrity, verifiability.
5.26	Grounding of <i>Sichem Osprey</i>	Anti-collision radar thresholds were apparently set incorrectly; there were also sizeable discrepancies between positions plotted on a chart and those displayed on the radar. The vessel grounded at more than 16 knots; no pollution occurred.	Maritime	2010	Integrity, accuracy
5.27	Near Collision of Trains, Cootamundra	Following a signalling system design error, a passenger train had to unexpectedly apply its brakes; it stopped just 5 m short of a goods train.	Rail	2009	Integrity, completeness
5.28	Cedars-Sinai Medical Center Scanner	A software misconfiguration led to 206 patients receiving radiation doses approximately eight times higher than intended; the error persisted for 18 months.	Medical	2008	Verifiability
5.29	Grounding of <i>The Pride of Canterbury</i>	An unapproved electronic chart system (ECS) was apparently being used as the primary means of navigation for the passenger ferry <i>The Pride of Canterbury</i> . Due to user settings a charted wreck would not have been displayed on this system. The vessel grounded on the wreck, causing severe damage to her port propeller system.	Maritime	2008	Accuracy, completeness, intended destination / usage
5.30	LOT Flight 282	Incorrect data input to the Flight Management System, 'E' rather than 'W', meant loss of instruments. Aircraft had to return to London Heathrow airport (LHR).	Air	2008	Integrity, fidelity / representation
5.31	Annabella container ship – Baltic Sea	Software developed loading plan using incorrect container specifications	Maritime	2007	Integrity, verifiability, fidelity / representation
5.32	Comair Flight 5191	Inaccurate (out-of-date) aerodrome charts led to take-off being attempted from the wrong runway. Aircraft overran the runway; 49 fatalities.	Air	2006	Timeliness, completeness

Continued on next page

Table 5.2: Incidents and accidents (continued)

Ref.	Title	Summary	Domain	Year	Data Property
5.33	Überlingen Mid-Air Collision	Contradictory advice from traffic collision avoidance system (TCAS) and an air traffic controller led to a mid-air collision between two TCAS -equipped aircraft. 71 fatalities.	Air	2002	Consistency, availability, timeliness
5.34	Fort Drum Artillery Incident	Movement of an artillery site led to errors in targeting. Artillery shells were fired more than a mile off target: 2 soldiers killed; 13 injured.	Defence	2002	Integrity, verifiability
5.35	Early release from Washington State Prison	A software update led to miscalculation of the time an inmate was due to serve in prison. Although the results of the calculation could easily be checked, the problem persisted for 13 years and over 2,000 offenders were released early.	Policing	2002	Integrity, verifiability
5.36	Mars Climate Orbiter	A mismatch in the units used by two software teams led to errors in the Flight Management System and, ultimately, the loss of a multi-million dollar space mission.	Space	1998	Consistency
5.37	Crash into Nimitz Hill, Guam	Controlled flight into terrain (CFIT) ; the ground-based minimum safe altitude warning (MSAW) designed to alert air traffic controllers had been inhibited. 228 fatalities; 26 serious injuries.	Air	1997	Continuity, fidelity / representation
5.38	San Bernardino derailment and pipeline rupture	Criticality of train weight not recognized, resulting in multiple fatalities	Rail	1989	Integrity, completeness, accuracy, timeliness, verifiability, fidelity / representation, lifetime
5.39	Lake Peigneur Drilling Accident	While drilling a test well, a rig crew inadvertently caused a flood in a nearby salt mine. The previously freshwater lake became a salt water lake and the flow of a river was reversed.	Oil and Gas	1980	Verifiability

5.2 Post Office Horizon System

Post Office Limited is a company wholly owned by the UK government, and provides a variety of counter services to the general public. These include postal services, banking services including currency exchange, issue of international driving permits, driving licence renewals, passport application checks, benefits payments, and various other services. A small number of branches are operated by Post Office Limited

itself (“Crown Post Offices”), but the vast majority are operated under contract (“Branch Post Offices”) by independent persons known as [subpostmasters \(SPMs\)](#).

In the 1990s, [Post Office Counters Limited \(POCL\)](#) (the name at that time for what would become Post Office Limited), the Department of Social Security (the government department at that time responsible for the Post Office) and ICL agreed to replace the paper-based accounting scheme at Post Office branches with an electronic system, in particular to allow the payment of benefits by electronic transfer instead of cash. A pilot system known as Pathway was rolled out to a small number of branches in 1996, but was subsequently abandoned due to “greater than expected complexity”. However, [POCL](#) and ICL decided to continue with a system based on Pathway to automate branch post offices. ICL was acquired by Fujitsu in 1998, and the resulting system, known as Horizon (now known as Legacy Horizon), was rolled out from 1999 and a version that combined management accounting functions and electronic point-of-sale functions, Horizon Online, was rolled out from 2010.

From the outset, Horizon has been a [data-driven system](#) “in which any requirement which might change frequently is encoded as data, rather than software code. The code is written and tested to work with all allowed values of the data” [11].

Shortly after the introduction of (legacy) Horizon, there was a sharp increase in [SPMs](#) reporting accounting shortfalls; the products that Horizon showed they had sold far exceeded the money the [SPMs](#) had taken. However, unlike the paper system, Horizon did not allow [SPMs](#) any access to the transaction records, so they were unable to trace the cause of the discrepancy. Under the terms of their contract with Post Office Limited the [SPMs](#) were obliged to make good any shortfall unless they could prove they were not at fault. Without access to the accounting trail there was no possible way for them to do that.

Following the roll out of Horizon, Post Office Limited “prosecuted more than 700 [SPMs](#) for crimes such as theft and false accounting. Hundreds of [SPMs](#) were sent to prison and many more received punishments such as being forced to do community service and having to wear electronic tags. [...] Hundreds were made bankrupt, losing their livelihood, and many struggled after being forced to pay the Post Office to cover shortfalls that didn’t exist outside the Horizon system. The lives of the victims and their families were severely impacted, with several suicides linked to the scandal and many cases of illness caused by stress.” [12]. The shortfalls were eventually found to have been due to faults in the Horizon system and to Fujitsu staff changing the accounts, apparently on the instruction of the Post Office, without the knowledge of the [SPMs](#).

The false prosecutions resulting from the faults in the Horizon system are at the time of writing subject to a public inquiry. The [Criminal Cases Review Commission \(CCRC\)](#) described the prosecutions as “the most widespread miscarriage of justice the [CCRC](#) has ever seen and represents the biggest single series of wrongful convictions in British legal history” [13].

The data issues included:

- Neither [SPMs](#) nor Post Office Limited were able to access the full data necessary to identify the source of accounting discrepancies [14, §995] and in particular [SPMs](#)’ “ability to investigate was itself similarly limited. The expert agreement [...] makes it clear in IT terms (based on the transaction data and reporting functions available to [SPMs](#)) that [SPMs](#) simply could not identify apparent or alleged discrepancies and shortfalls, their causes, nor access or properly identify transactions recorded on Horizon, themselves. They required the co-operation of the Post Office [14, §1000]”.
- Change control processes for the data representing the products and services provided were inadequate [11, §54].

- Transactions within Horizon were not atomic, so transactional *integrity* was not maintained: if a transaction failed, the payment for the goods or service could be recorded without showing on the SPM's point of sale system. In that case SPMs were instructed to retry the transaction, so the payment would be recorded multiple times for a single transaction. Possible causes of failure of a transaction included the speed with which a button on the point-of-sale terminal was pressed [14, §113].
- Horizon did not maintain correct double-entry bookkeeping even within transactions that were completed normally [11, §128ff]
- Fujitsu, apparently on Post Office Limited's instruction, changed accounting transactions without the knowledge of the SPMs responsible for those transactions [11, §61.4].
- Records were not kept of the occasions accounting transactions were altered by Fujitsu [14, §1013, §1014]
- Post Office Limited staff were given unnecessary top-level security access to the accounting data [14, §390].

5.3 Battle of Agincourt

At the battle of Agincourt (1415), the French cavalry had little regard for the English longbowmen. They appear to have been so proud that they thought little of the power of the English archers. The French underestimated the longbow at Agincourt in several ways. Thomas Walsingham, an English chronicler, described how the French were “utterly astonished” by the effectiveness of the English longbowmen, noting that “the French never thought that such weapons could have such power”. This chronicler, among others, demonstrates that the French underestimated the longbow at Agincourt due to their misconceptions about its range, *accuracy*, and impact on their cavalry. This underestimation ultimately contributed to their defeat at the hands of the English army.

The French assessment of English capability was probably based on their own experience with crossbows, which had more limited range and effectiveness. Thus even when Henry moved his troops closer to the French army, so as to be within longbow range, the French still did not perceive the threat.

Data properties involved: Their assessment was based upon incorrect data, resulting in loss of *integrity* and *completeness*. Although they understood the principles of one type of bow, the French data could be regarded as inaccurate, due to their read-across from one type of weapon to another, resulting in loss of the *accuracy* and the *fidelity / representation* property. However another view of the problem was that the French used data from what (to them) was old technology – the crossbow – and failed to keep their data up to date as technology advanced, leading to loss of the *timeliness* property.

Links

- Agincourt: <https://www.longbow-archers.com/historyagincourt.html> (Accessed 11 February 2024)
- Walsingham, Historia Anglicana, Volume II: <https://archive.org/details/thomwalsinghamh00walsgoog/page/312/mode/2up?view=theater> (Accessed 11 February 2024)

5.4 Gemini V

The Gemini V crewed spaceflight took place in 1965. It lasted eight days – twice as long as Gemini IV, thereby demonstrating that a spaceflight long enough to get to the moon and back was feasible.

There had been various system failures during the mission. However during re-entry, systems behaved correctly, and the crew were able to control the descent as planned. However the spacecraft landed eighty miles short of its intended landing zone.

Post-mission analysis revealed that the computing equipment had not failed. The error had been introduced by a simple **data error** – an assumption that the Earth turns 360° in 24 hours. The 24 hour day was based upon the position of the Sun at midday, and a simplistic understanding of Earth's rotation suggests that the ball must have rotated 360° for the Sun to appear in the same vertical plane. However the Earth follows an orbit around the Sun, making one full orbit per year. Thus on average, every 24 hours, the Earth moves $360 \div 365.24 = 0.99^\circ$ along its orbit, and must over-rotate by a similar amount for the Sun to appear in the same place. In other words, whilst everybody “knows” that the Earth rotates 360° per day, that figure is incorrect.

As the computer had been given a parameter of 360° , instead of the accurate value, the calculations to reach the landing site were inaccurate.

Data properties involved: The approximation was a loss of the **accuracy** property. Its introduction resulted from a lack of consideration of celestial movement, and so could potentially also be regarded as a loss of both the **traceability** and **fidelity / representation** properties.

Links

- A general outline of the Gemini V mission: https://en.wikipedia.org/wiki/Gemini_5 (Accessed 26 January 2022)
- An explanation of how Earth's orbit results in more than 360° rotation in 24 hours: https://en.wikipedia.org/wiki/Sidereal_time (Accessed 26 January 2022)
- The Gemini V mission report: <https://www.ibiblio.org/apollo/Documents/Gemini5MissionReport.pdf> (Accessed 26 January 2022)

5.5 What3Words

The What3Words application is intended to provide a means to easily and unambiguously define a location anywhere on the surface of the Earth. The ease with which the application can be used has led to its use by emergency services worldwide, for locating people in need of assistance – in safety terms, it has provided a useful form of **mitigation**, enabling emergency services to easily locate people, whether simply at the roadside, or somewhere on a mountainside. However there have been a number of claims that What3Words is not suitable for this purpose.

According to the Telegraph (1 June 2021), Mountain Rescue England and Wales (MREW) claimed it had been told to go to 45 locations in the past 12 months which had turned out to be incorrect, whilst a rescue request that indicated a location in Australia actually emanated from a location in China. The article suggested that the problem was a combination of local accents and spelling errors.

However the BBC reported that research by Andrew Tierney indicates that the problem is more systemic, and reported:

- The dictionary used by What3Words includes a number of homophones – words that have the same pronunciation, but different spelling.

- Similar sounding words
- Plurals

Cybergibbons illustrated the problem by presenting a list of thirty two pairs of words in the What3Words dictionary that appear to be extremely similar in pronunciation. They also pointed out a further algorithmic deficiency – that plural words can be followed by a word beginning with the letter “s”. Their examples include likely.stage.sock and likely.stages.sock, which denote locations on opposite sides of the River Clyde, and illustrate how in mountain rescue situations, even a small error in location could result in a serious delay to emergency provision.

Thus it appears that the system is working as designed, by providing a simple means to carry out geolocation. However a tool that was designed as a social application to help friends meet up has now been applied to a safety-related domain, without being re-engineered for that more demanding purpose. Three words will appear on the screen of the person requiring help, and if those same three words are used by the emergency services, the parties will be able to meet up. However that transfer is carried out by voice, where ambiguity can introduce error. The dictionary appears to contain homophones, plurals are used, and the algorithm permits the selection of pairs of words where the ending of one word cannot easily be distinguished from the start of the following word – each of these failings can lead to errors in communication.

Data properties involved: The ambiguities can lead to loss of *integrity*, whilst the resulting distortion of world view is a loss of *consistency*, and the Data Issue is one of *Aliasing*.

Links

- The Telegraph article: <https://www.telegraph.co.uk/news/2021/06/01/rescuers-directed-china-australia-what3words-app-regional-accent/> (Accessed 16 January 2022)
- A BBC article: <https://www.bbc.co.uk/news/technology-56901363> (Accessed 16 January 2022)
- Cybergibbons report: <https://cybergibbons.com/security-2/why-what3words-is-not-suitable-for-safety-critical-applications/> (Accessed 16 January 2022)

5.6 Java log4j Library Vulnerability

On Dec 10th 2021 a new critical zero-day vulnerability was detected that affected Apache Log4j 2 Java library. It adversely impacted the digital domain and security systems worldwide.

The vulnerability, when exploited, permitted remote code execution on the vulnerable server with system-level privileges.

Log4j is a highly configurable logging mechanism for Java (“log4j”) that is used for documentation and debugging. Although originally developed for the Apache web server, it has been used part of many commercial applications, including network monitoring tools and even games such as Minecraft.

The exploit was a combination of the Java code that contains different logging functions (typically `error()`, `warn()`, `info()`, `debug()`, ...) and a configuration file. The configuration file specifies which **information** shall be added to the log-file, the associated format, and how to “interpret” the logged data.

The security risk was that the logging mechanism was by default configured in a way such that it interpreted the logged data, and that the logged data that the user entered could be used to attack the server. For

example if a user were to enter into the name field of a html-page instead of his name a “delete *.*” command, along with certain escape sequences, it might cause huge damage on the server – if this data were logged from the software and interpreted from the configuration file.

Data property involved: *integrity*.

Links

- Apache provides details on security issues with the log4j library, including available fixes, on its website <https://logging.apache.org/log4j/2.x/security.html> (Accessed: 09/01/2022)
- Further details may be found at <https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2021-44228> (Accessed: 09/01/2022)

5.7 Immensa False Negative Covid-19 Tests

Failings at the Immensa lab in Wolverhampton led to an estimated 43,000 people with Covid-19 being mistakenly given negative PCR results; they thought they were in the clear, but were actually positive for Covid-19. This contributed to soaring rates of infection across the South West and Wales.

It took the government almost a month to identify the issue and to stop sending PCR tests there.

It is unknown how much the virus spread in that time, but the effects of this mismanagement are potentially huge. Professor Deepti Gurdasani, a senior lecturer in epidemiology at Queen Mary University, estimates that the false negatives may have caused up to 200,000 further Covid-19 infections, and more than 1,000 avoidable deaths.

Data properties involved: *integrity*, *accuracy* and possibly *traceability* and *verifiability*.

Links

- BBC programme Inside Science containing discussion with Professor Gurdasani: <https://www.bbc.co.uk/programmes/m0010q9x> (Accessed: 09/01/2022)
- <https://goodlawproject.org/news/immensa-update/> (Accessed: 09/01/2022)
- <https://www.gov.uk/government/news/testing-at-private-lab-suspended-following-nhs-test-and-trace-investigation> (Accessed: 09/01/2022)
- <https://www.theguardian.com/world/2021/dec/21/immensa-lab-month-delay-before-incorrect-covid-tests-stopped> (Accessed: 09/01/2022)

5.8 Covid-19 Test Results Silently Deleted by Excel

In early October 2020 the British Government announced that 15841 positive Covid-19 test results had not been reported in the totals for England between 25 September and 2 October. This also meant that the contacts of those people were not traced, or asked to self-isolate, meaning that the virus might have spread further than it would otherwise have done, and possibly have taken additional lives.

The results were silently discarded as they were imported into the Public Health England [database](#). Results were delivered as a [comma separated variable \(CSV\)](#) file, which was imported into an Excel spreadsheet

“template”, using the “.xls” format, which was then in turn imported into the national [database](#). The “.xls” format has a limit of 65536 (2^{16}) rows, and rows beyond this limit in the [CSV](#) file were silently discarded. (Data category: dynamic; properties lost: [integrity](#), [completeness](#), [timeliness](#), [availability](#), [fidelity / representation](#))

The newer “.xlsx” file format would have increased the row limit to 1048576 (2^{20}) rows before suffering the same problem, but the [CSV](#) file format has no limit on the number of rows.

This demonstrates the danger of using [COTS](#) software for safety-related functions without fully analysing its limitations. A decision had already been made to replace this system, but had not been acted upon. It is also an example of dark data, where it comes under the *data we don't know are missing*: “unknown unknowns”, and the *Missing What Matters* categories.

Finally, it is an example of where an error is known to the system, but not reported (adequately) to the user (Data Category: dynamic; properties lost: [timeliness](#), [availability](#)).

Links

- <https://www.bbc.co.uk/news/technology-54423988> (Accessed: 11/01/21)
- <https://www.bbc.co.uk/news/uk-54422505> (Accessed: 12/01/21)

5.9 Boeing 737 MAX 8 Crashes

On October 29th 2018, Lion Air flight 610 was lost with all on board when it flew into the sea. On March 10th 2019, Ethiopian Airlines flight 302 flew into the ground. In each case the aircraft was a Boeing 737 MAX 8 – the latest modernised iteration of the 737 airframe design, and in each, a software system called the [manoeuvring characteristics augmentation system \(MCAS\)](#) has been established as the principle cause of the crashes.

The [MCAS](#) system was introduced because bigger, and hence more fuel efficient, engines are used on the 737 MAX 8, which had to be positioned higher on the wing and further forward (to ensure sufficient ground clearance for the larger engines). This changed the aerodynamic properties of the aircraft and meant the 737 MAX 8 tended to pitch up during high angles of attack. [MCAS](#) was introduced to counter this effect by taking input from an [AoA](#) sensor and commanding the horizontal stabiliser control surfaces to bring the nose down. Although the aircraft has two [AoA](#) sensors, only one sensor gave input to [MCAS](#) at any one time, meaning that a single sensor failure could cause the nose to be forced down incorrectly. The design of [MCAS](#) allowed this to happen repeatedly, which if left unchecked would eventually force the aircraft into an unrecoverable dive.

[MCAS](#) did not compare data from the two sensors in order to detect a discrepancy between them, and hence indicate that the sensor data was not trustworthy (data category: dynamic data; properties lost: [integrity](#), [accuracy](#), [fidelity / representation](#); [mitigations](#) unused: redundancy). Furthermore, the function (outside MCAS) to report a discrepancy between the sensors to the pilots was not enabled, reducing the crew's ability to respond appropriately (data category: dynamic; properties lost: [availability](#)).

Several other data-safety-related failures can also be found in the report on flight 610:

- The [MCAS](#) system was not described in the pilot's manual and training materials (Data Category: Staffing, Properties lost: [availability](#))

- There was no indication to the pilots that MCAS was active (Data Category: Dynamic; Properties lost: *availability*).
- The AoA sensor fitted to flight 610 was incorrectly calibrated during a previous repair, reporting an angle 21° higher than the correct value. But this was not detected during the repair (Data Category: Justification; Properties lost: *availability, fidelity / representation, verifiability*).
- The evidence of the testing of the AoA sensor after fitting to flight 610 by the maintenance crew was erroneous (Data Category: Justification; Properties lost: *integrity, availability, verifiability*).
- 31 pages were missing from the maintenance log-book for flight 610, including the records of the testing of the AoA sensor after fitting to the aircraft (Data Category: Asset; Properties lost: *completeness, availability, verifiability*).
- During the previous flight of the aircraft on flight 610, the pilots experienced repeated activation of the stick shaker, and other alarms, which were caused by the faulty AoA sensor. However they did not fully report all the issues in the flight logs, and so the maintenance crew's remediation activities did not lead them to suspect an issue with the sensor. (Data Category: Performance, Properties lost: *availability*).
- During flight testing, *adaptation data* was changed to give MCAS more authority, without a new safety impact assessment (Data Category: Adaptation; Properties lost: *verifiability*).

Links

- http://knkt.dephub.go.id/knkt/ntsc_aviation/baru/2018-035-PK-LQP-Final-report.pdf (Accessed: 20/12/19)
- https://en.wikipedia.org/wiki/Boeing_737_MAX_groundings (Accessed: 20/12/19)
- https://en.wikipedia.org/wiki/Lion_Air_Flight_610 (Accessed: 20/12/19)
- https://en.wikipedia.org/wiki/Ethiopian_Airlines_Flight_302 (Accessed: 20/12/19)
- <https://www.businessinsider.in/thelife/boeing-reportedly-made-the-flight-control-system-that-mistakenly-activated-during-2-deadly-crashes-4-times-stronger-while-creating-the-737-max/articleshow/68840690.cms> (Accessed: 03/01/20)
- <https://www.theverge.com/2019/5/2/18518176/boeing-737-max-crash-problems-human-error-mcas-faa> (Accessed: 04/01/20)
- The design, development & Certification of the Boeing 737 MAX, September 2020, The House Committee on Transportation & Infrastructure.
<https://transportation.house.gov/committee-activity/boeing-737-max-investigation> (Accessed 22/01/2021)

5.10 Loss of Soyuz-2.1b Rocket Carrying Meteor-M 2-1 Weather Satellite

On November 28th, 2017, the second launch took place from Russia's new launch site at Vostochny, carrying the Meteor-M No.2-1 polar-orbiting weather satellite, and 18 small satellites flying as secondary payloads. The launch appeared successful, but several hours later it was announced that it had not been possible to

establish communications with the weather satellite, because it was not in its target orbit. An unconfirmed report claimed that the rocket was in the wrong orientation during its initial burn, and crashed into the Atlantic ocean. The following January, Russian deputy prime minister Dmitry Rogozin reported that the 2.6bn-ruble (\$45m) satellite was lost because the Soyuz-2.1b launch vehicle had been programmed to take off from Baikonur, and not from the actual launch site at Vostochny. This is an example of a problem with data in the Adaptation category, which did not map correctly to the real-world entity that it was modelling, and hence lost the *fidelity / representation* property. Other properties that were lost include *integrity* and *verifiability*.

Links

- <https://www.theguardian.com/world/2017/dec/28/russian-satellite-lost-wrong-spaceport-meteor-m>
(Accessed: 14/01/20)
- <https://www.space.com/39270-russian-weather-satellite-doomed-human-error.html>
(Accessed: 22/01/20)
- <https://www.space.com/38918-russian-satellites-lose-contact-after-launch.html>
(Accessed: 22/01/20)

5.11 Cambrian Line Data Loss

In the United Kingdom, railway signalling is primarily implemented through trackside signals. In 2011, a trial system was installed in the Machynlleth signalling control centre, which enabled suitably equipped trains travelling over the part of the rail network that it controlled (the Cambrian lines) to acquire data relating to speed restrictions and display these to the driver. The Cambrian lines are a collection of rail tracks which run along the Welsh coast and as far inland as Shrewsbury. On 20th October 2017, a driver reported that his train had failed to display speed restriction data.

The initiating event was just after 23:00 hrs on 19th October 2017, when a train automatically requested a movement authority (permission to travel on a specified part of the rail network) which had already been allocated to another train. This error condition is known to occur several times per year and causes an automatic software reset to be invoked. A well-established set of processes are used by the signalling centre staff to return the rail system to normal operation. The staff followed their processes, and once the system was functional, allowed the final three trains of the day to complete their journeys. On the following day, it was the driver of the fourth train of the day who noticed the error and reported the failure.

Subsequent investigation revealed that speed restriction data had been unavailable since the software reset, resulting in six trains completing their journeys without that data, before the driver of the seventh train observed the problem.

The most significant risk identified thus far by the rail Accident Investigation Board is that a number of the speed restrictions which were in place on the Cambrian lines had been invoked to allow pedestrians at level crossings sufficient time to take action, when observing an approaching train. Luckily the failure did not result in an accident – but this was due to luck, not fail-safe systems on the railway.

This incident highlights the importance of the *availability data property*, as the data had been silently unavailable to trains. In addition, much of the digital audit trail relating to the failure was lost during repeated attempts to correct the problem and get the rail network running again, a loss of the *history data property*.

Links

- Interim report: https://assets.publishing.service.gov.uk/media/5bc871d5e5274a0956564a41/Ir012018_181018_Cambrian_TSrs.pdf (accessed 31 December 2018).
- Final report: <https://www.gov.uk/government/news/report-172019-loss-of-safety-critical-signalling-data-on-the-cambrian-coast-line> (accessed 14 January 2020).

5.12 Loss of Irish Rescue Helicopter

On 14 March 2017, an Irish [search and rescue \(SAR\)](#) helicopter apparently suffered [CFIT](#); all four crew members were killed. *Note: At the time of writing the investigation is continuing. The following discussion is based on a preliminary report from the Air Accident Investigation Unit, Ireland.*

The [SAR](#) helicopter was responding to a medical emergency on board a fishing vessel. It left Dublin and requested a route to Blacksod to refuel. The flight data recorded on the [health and usage monitoring system \(HUMS\)](#) showed the helicopter was in stable level flight until the final few seconds, when it pitched up rapidly and impacted with terrain at the western end of Black rock.

The helicopter was equipped with an [enhanced ground proximity warning system \(EGPWS\)](#). This is designed to decrease instances of [CFIT](#) by increasing pilot situational awareness, including the use of alerts and warnings: it is not intended to be used for aircraft navigation. The [EGPWS](#) can provide terrain alerting using look ahead algorithms, which take [information](#) from the aircraft (e.g. position, attitude, heading) and use this in conjunction with internal terrain and obstacle [databases](#). Neither the Black rock lighthouse nor the island's terrain were included in the [EGPWS databases](#); these [databases](#) had been sourced from external suppliers by the [EGPWS](#) manufacturer.

The preliminary investigation also notes that the flight crew were following an operator-specific route guide: a review of such guides has been recommended.

This incident potentially illustrates the importance of the [fidelity / representation](#) and [completeness data properties](#), with respect to the [EGPWS database](#), and the [accuracy data property](#), with respect to the route guides.

Links

- <http://www.aaiu.ie/sites/default/files/report-attachments/rEPOrT%202017-006%20PrELIMINArY.pdf> (accessed 29 November 2017).

5.13 Loss of Schiaparelli Mars Lander

The Schiaparelli module, also known as the [entry demonstrator module \(EDM\)](#), was part of the [European Space Agency \(ESA\)](#)'s ExoMars 2016 mission. The objective was to validate and demonstrate entry, descent and landing on Mars in preparation for the ExoMars 2020 mission.

On 19 October 2016, the [EDM](#) entered the Mars atmosphere at 14:42:07 (UTC). During its entry and descent it constantly transmitted telemetry. Its signal was lost at 14:47:22 (UTC), about 43 seconds before expected

touchdown. On 20 October, a camera on NASA's Mars reconnaissance Orbiter imaged the planned landing site and observed crash debris.

During entry, a parachute was deployed as planned. This triggered oscillations that saturated the IMU. Integration of this saturated value caused a significant error in predicted attitude. As the descent continued, a radar Doppler altimeter (RDA) was turned on. The significant attitude error led to large discrepancies between the IMU and the RDA. The nature of the guidance and navigation control software meant that this discrepancy led to a premature declaration of touchdown. As such, the parachute was jettisoned too early, causing the EDM to crash into the planet's surface.

The investigation determined the rates that saturated the IMU could have been predicted. Limitations in the modelling of parachute dynamics meant they were not. The investigation also noted issues with the persistence of the flag used to denote IMU saturation, as well as inadequate handling of this saturation by the guidance software.

This incident illustrates the importance of the *fidelity / representation data property*, with respect to the modelling, and the *integrity data property*, with respect to the persistence time of the saturation flag.

Links

- <http://exploration.esa.int/mars/59176-exomars-2016-schiaparelli-anomaly-inquiry/> (accessed 29 November 2017).

5.14 Interception of Communications

In July 2015, it was reported that a public authority was undertaking an investigation into the uploading of indecent images of children and requested details of the account connected to the internet protocol (IP) address used to upload the images. Issues with a new upgrade of the communication provider's system resulted in incorrect data being disclosed. Investigations revealed that a further five requests had resulted in incorrect data being disclosed. Data was acquired in six cases that related to individuals unconnected with the investigations. In one of these cases a welfare check was delayed on a child believed to be in crisis.

Under the Regulation of Investigatory Powers Act 2000, internet service providers and indeed other communication service providers (e.g. mobile phone network providers) are required to provide data to investigatory bodies such as the police. This data can be used to support criminal investigation and prosecutions and in the protection of vulnerable children and adults. The data clearly has the potential to be safety related, but there is no obligation for data providers to treat it as such. In this case the data errors led to a child being exposed to additional risk of harm.

This incident highlights the importance of the *integrity data property*. It also shows the applicability of data safety guidance to areas that are not traditionally encompassed by safety engineering.

Links

- [https://www.ipco.org.uk/docs/iocco/2015%20Half-yearly%20report%20\(web%20version\).pdf](https://www.ipco.org.uk/docs/iocco/2015%20Half-yearly%20report%20(web%20version).pdf) (accessed 9 January 2019).

5.15 A400M, Torque Calibration Parameters

On 9 May 2015, just minutes into a routine, pre-delivery test flight an Airbus A400M military plane crashed in Spain, killing four of the six crew. Three of the four engines had become stuck at high power and initially did not respond to the crew's attempts to control the power setting in the normal way. Pilots then succeeded in reducing power only after selecting the thrust levers to idle. The engines subsequently remained stuck in this mode. In an attempt to return to the airport, the aircraft struck power lines and crashed.

Although not confirmed, reports suggest the torque calibration parameters for the engines were wiped during a software installation. The torque calibration data is needed to measure and interpret [information](#) coming back from the A400M's engines, and is crucial for the [electronic control units \(ECUs\)](#) that control the aircraft's power systems.

This accident highlights the importance of the [completeness data property](#), specifically with respect to the torque calibration parameters.

Links

- <http://www.bbc.co.uk/news/technology-33078767> (accessed 29 November 2017).
- <http://www.reuters.com/article/us-airbus-a400m-idUSKBN0OP2AS20150609> (accessed 29 November 2017).

5.16 RN Submarine, Trawler Karen

On 15 April 2015, a submerged royal Navy submarine snagged the fishing gear of the UK registered trawler *Karen*, 15 miles south-east of Ardglass, Northern Ireland. *Karen* had been trawling for prawns on a westerly heading at 2.8 knots when its fishing gear was snagged and it was dragged backwards at about 7 knots. *Karen*'s crew managed to release both winch brakes, freeing the trawl warps; the starboard warp ran out completely but the port warp became fouled on the winch drum, causing the vessel to heel heavily to port and its stern to be pulled underwater. *Karen* broke free from the submarine when the port warp parted; there was structural damage to the vessel but it returned to Ardglass safely under its own power. Evidence of the collision on board the submarine was either not observed or misinterpreted.

The nature of sub-surface operations requires the use of sonar technology to detect collision hazards. Detection in this way is reliant on noise emanating from contacts. In this instance the fishing trawler was detected but misidentified as a merchant vessel rather than a fishing vessel because the submarine's sonar operators did not detect or report hearing trawl noise. Given the number of vessels operating in the area, it is almost certain that the noise levels being generated would have been extremely high, with noise from one vessel masking the noise from another. Such a situation would make it very difficult for the operators to methodically identify and analyse each contact, in particular to identify discrete acoustic classification clues such as trawl noise. As a result the trawler was assessed to be that of a small merchant vessel and the command team's perception would have been that no risk of collision could exist between a submarine at safe depth and a merchant vessel.

review concluded that the submarine was operating near to the limit of its capability. Given that all the submarine's systems were reported to be functioning properly, it was apparent that the submarine's limit of capability had, in reality, been exceeded, with its sonar and command teams becoming cognitively overloaded, leading to degraded situational awareness and poor decision making.

In conclusion, the Maritime Accident Investigation Board (MAIB) report stated, "The collision happened

because the submarine's command team believed *Karen* to be a merchant ship, so they did not perceive any risk of collision or need for avoiding action."

This incident highlights the importance of the [resolution data property](#), specifically with regards to resolving a trawler and a merchant vessel. It also highlights the importance of the [integrity](#), specifically with regards to the [information](#) provided from the sonar team to the command team.

Links

- <https://www.gov.uk/maib-reports/collision-between-the-stern-trawler-karen-and-a-dived-royal-navy-submarine> (accessed 29 November 2017).

5.17 Turkish Airlines A330

During March 2015, an Airbus A330-303, operated by THY Turkish Airlines, suffered a runway excursion accident upon landing at Kathmandu-Tribhuvan (KTM).

Flight TK726 was a regular passenger service from Istanbul-Atatürk (IST) to Kathmandu, Nepal. The flight was the first international flight to arrive that morning. After descending from cruising altitude, it entered a holding pattern. It was subsequently cleared for a [VHF omnidirectional range \(VOR\) / distance measuring equipment \(DME\)](#) approach to runway 02.

This approach was abandoned at about the Missed Approach Point at 1DME and the aircraft performed a go around. The aircraft circled and positioned for a second approach to runway 02. The aircraft touched down to the left of the runway centre line with the left hand main gear off the paved runway surface. It ran onto soft soil and the nose landing gear collapsed. Following the accident the aircraft was written off.

The aircraft touched down to the left of the centreline because the [flight management guidance system \(FMGS\)](#) navigation [database](#) contained threshold coordinates for a proposed displacement of the runway 02 threshold. This was later withdrawn through a [notice to airmen \(NOTAM\)](#), but had not been updated by the airline in the [database](#). Additionally, the coordinates that were initially published were inaccurate, causing the threshold coordinates to be offset to the left of the actual threshold. This had been noticed and reported by a previous Turkish Airlines flight on March 2. The changes had not been performed by the time TK726 landed at Kathmandu.

Among the safety recommendations stated in the accident report were:

- "The operator must ensure that the correct navigation data are uploaded on flight management guidance system";
- "The operator should establish a system of verifying the quality of charts prepared by the service provider";
- "The operator should establish a system of checking the validity of the [flight management system \(FMS\) database](#)"; and
- "Civil Aviation Authority of Nepal must ensure that raw [aeronautical information](#) / data are provided by the aerodrome authorities taking into account of its accuracy and [integrity](#) requirements for [aeronautical data](#) as specified by [International Civil Aviation Organization \(ICAO\)](#) Annex 15 and its Aeronautical Information Service Manual."

This incident highlights the importance of three [data properties](#): [accuracy](#); [timeliness](#); and [verifiability](#). All of these apply to data describing the runway's location.

Links

- http://www.tourism.gov.np/downloadfile/TURKISH_AirLINE_Final_report_finalcopy4.pdf (accessed 29 November 2017 – no longer available from this location, but on 9 January 2019 was still available through archive.org).

5.18 Dallas Hospital Ebola Incident

On 26th September 2014, a Dallas hospital mistakenly sent home a man who had the Ebola virus having missed what would have appeared to be an obvious potential case: a Liberian citizen with fever and abdominal pain who said he had recently travelled from Liberia. The man later returned to the hospital, was eventually diagnosed with the illness, but subsequently died. Two nurses that had treated the man also contracted the virus but later recovered.

There have been mixed reports on the cause of the problem, but it is clear that external social phenomena such as the Ebola outbreak, which are outside the hospital's [electronic health record \(EHR\)](#) system and processes, can change the safety significance of data held in the [EHR](#). If the importance of the data is not recognized and elevated appropriately in the support tools and processes, then the risk of unintended harm can increase. This conclusion is reinforced by system vendors who subsequently updated their systems to reflect the Ebola crisis in light of the Dallas incident.

This incident highlights the importance of the [completeness](#) and [format data properties](#), in that [information](#) about the Ebola outbreak was apparently either not available, or not available in a usable form, to decision makers.

Links

- <http://www.nbcnews.com/storyline/ebola-virus-outbreak/texas-hospital-makeschanges-after-ebola-patient-turned-away-n217296> (accessed 29 November 2017).

5.19 Qantas Boeing 737 Take-Off

On 1 August 2014, a Qantas Boeing 737-838 aircraft commenced take-off from Sydney Airport, New South Wales. The flight was a scheduled passenger service from Sydney to Darwin, Northern Territory.

While the aircraft was climbing to cruise level, a cabin crew member reported hearing a “squeak” during rotation. Suspecting a tail strike, the flight crew conducted the tail strike checklist and contacted the operator's maintenance support. With no indication of a tailstrike, they continued to Darwin and landed normally. After landing, the captain noticed some paint was scraped off the protective tailskid. This indicated the aircraft's tail only just contacted the ground during take-off.

The [Australian Transport Safety Bureau \(ATSB\)](#) found the tail strike was the result of two independent and inadvertent data entry errors in calculating the take-off performance data. As a result, the take-off weight used was 10 tonnes lower than the actual weight. This resulted in the take-off speeds and engine thrust setting calculated and used for the take-off being too low. Hence, when the aircraft was rotated, it overpitched and contacted the runway.

The [ATSB](#) also identified that the Qantas procedure for conducting a check of the Vref40 speed could be misinterpreted. This negated the effectiveness of that check as a defence for identifying data entry errors. In this case, uncorrected errors affected the [integrity](#) of the data used to calculate take-off parameters.

This incident highlights the importance of the [integrity](#) and [verifiability data properties](#), with respect to the data used to calculate take-off performance data.

Links

- http://www.atsb.gov.au/publications/investigation_reports/2014/aair/ao-2014-162.aspx (accessed 29 November 2017).

5.20 Qantas Boeing 737 Loading

On 9 May 2014, a Qantas Boeing 737 was preparing for departure from Canberra to Perth. There were 150 passengers, 87 of which were primary school children. These children were all seated together at the rear of the cabin.

A ‘name template’ was completed by a travel agent on behalf of the school group. This group was travelling from Perth to Canberra and returning back to Perth. Despite being marked as mandatory, the “Gender Description” field in this template was left blank; options for this field were “Adult”, “Child” and “Infant”.

As per company procedures, two days before the Perth-Canberra leg of their journey this group was ‘advance accepted’ into the booking system. Since the fields recording the number of children and young passengers in the group were blank, the Customer Service Agent assumed all of the group were adults. No loading-related issues were experienced during this flight.

Two days before the return flight the group was again “advance accepted” as all adults. This meant they had all been assigned an ‘adult weight’ of 87 kg. They were checked in at Canberra Airport and assigned seats at the rear of the aircraft. During take-off the aircraft appeared nose heavy. Significant back pressure was required to rotate the aircraft and lift off from the runway. The aircraft exceeded the calculated take-off safety speed by about 25 kt. The aircraft rose at a higher initial climb speed than usual, but the crew did not receive any warnings. No further issues were experienced during the flight.

This incident demonstrates the importance of the [completeness](#) Data y (i.e., ensuring that the mandatory “Gender Description” field was completed) and the [fidelity / representation data property](#) (i.e., ensuring the calculated aircraft loads and balances reflect the real situation). It also illustrates some potential difficulties associated with the use of default data.

Links

- http://www.atsb.gov.au/publications/investigation_reports/2014/aair/ao-2014-088.aspx (accessed 29 November 2017).

5.21 Grounding of Navigator Scorpio

On 3 January 2014, the liquefied gas carrier *Navigator Scorpio* ran aground on Haisborough Sand in the North Sea. The vessel was undamaged, no pollution occurred and after two and a half hours the vessel refloated on the rising tide.

The schedule for the *Navigator Scorpio* was changed close to the time of its departure. This change meant that additional North Sea coastal charts were required. These charts were delivered to the vessel shortly before its departure. However, they were not up to date with the latest corrections and they were not corrected prior to sailing. In addition, the passage plan (i.e., vessel route) was not checked by the master before sailing.

When the master checked the passage plan, which had been drawn up by the second officer (2/O), he suggested a change to a portion of the route. After discussion with the 2/O the route was left unchanged, but with a requirement that position fixes be obtained every five minutes rather than every fifteen. While acting as the sole bridge watch-keeper the 2/O was distracted by further passage planning activities and lost positional awareness. This led to the grounding of the vessel. After the grounding false [information](#) was added to the navigation chart to give the appearance that five minute positional fixes had been taken.

This incident highlights the importance of the [timeliness data property](#), with respect to both the additional North Sea charts and the master's check of the passage plan.

In addition, the fluidity of the chart data allowed the 2/O to make false post-grounding additions to create an incorrect impression. According to the MAIB's report, such actions are not uncommon. These actions affect the [verifiability](#) of the chart data, which makes post-accident investigations more complicated.

Links

- <https://assets.publishing.service.gov.uk/media/547c6f1740f0b6024100000d/NavigatorScorpio.pdf> (accessed 29 November 2017).

5.22 Loss of MQ-9 reaper

On 5 December 2012, an MQ-9 reaperMQ-9 reaper remotely piloted aircraft crashed in an unpopulated area three miles north-east of Mount Irish, Douglas County, Nevada. The crash occurred due to a stall, which was the result of an unrecognized reverse thrust condition. The aircraft and a number of pieces of ancillary equipment were destroyed. The total damage to United States government property was assessed at over \$9 million.

The investigation board concluded that the throttle settings of the [GCS](#) were incorrectly configured. This misconfiguration arose as the [GCS](#) was converted from supporting MQ-1 operations to supporting MQ-9 operations. It persisted despite the presence of a checklist, the completion of which should have identified the error. The misconfiguration meant that reverse thrust was commanded whenever the pilot's throttle was in any position except full forward.

This incident highlights the importance of the [consistency data property](#), with respect to the differences between the [GCS](#) settings and the aircraft it was meant to be controlling. It also highlights the importance of the [verifiability data properties](#), with respect to the [GCS](#) settings (and, in particular, the limitations of using checklists to verify data).

Links

- http://www.airforcemag.com/AircraftAccidentreports/Documents/2013/120512_MQ-9_Nevada_full.pdf (accessed 29 November 2017).

5.23 Boeing 737-33A at Chambéry Airport, France

On the 14 April 2012 and prior to departing Chambéry Airport in France, the crew of a Boeing 737 used an Electronic Flight Bag (EFB) computer to calculate the aircraft's take-off performance. During the use of the software application the commander omitted to input the aircraft's take-off weight and it defaulted to the previous flight's data. Compounding the issue was that none of the crew undertook a cross-check of the EFB's output and the pilot subsequently employed incorrect speed and thrust [information](#) for the take-off. The consequence of using the incorrect [information](#) was that the calculation of the required airspeed for rotation was too low and the pilot continued to increase the aircraft's pitch angle to the point whereby the tail hit the runway. There were no injuries sustained in this incident but the aircraft suffered damage.

Following its investigation, which examined the wider employment of computers to derive aircraft performance [information](#), the Air Accident Investigation Branch (AAIB) identified that there had been "a number" of previous accidents and incidents attributable to the "incorrect calculation of take-off performance"; and that due to the potential for degraded climb performance a catastrophic outcome could be envisaged. The AAIB also recognized that "take-off under-performance" is subtle and many other events of this nature may have been experienced but never reported. In its conclusions the AAIB acknowledged that using computers has "brought about improvements in [accuracy](#) and ease with which aircraft performance requirements can be made". However, there are "continued vulnerabilities" associated with the use of incorrect data that it is essential to control through "appropriately designed software and hardware". Although there were no injuries in this instance, this incident and the conclusions of the AAIB highlight some important points for "Safety-related [Information Systems](#)".

A clear chain of events was established that involved the use of incorrect [information](#) as a causal factor leading to an incident, which had the potential to be of a catastrophic nature;

- This was not an isolated incident;
- The crew did not appreciate the [criticality](#) of the EFB's [information](#) and it was used without [validation](#);
- The AAIB recognized the essential need for appropriate system development.

It is often recognized that data must be up to date, but the explicit need to prevent the use of old data can be omitted from the safety requirements. This incident illustrates the importance of the properties [timeliness](#), [suppression](#) and [lifetime](#).

Links

- *Air Accident Investigation Branch April Bulletin 4/2013* [on line] available at http://www.aaib.gov.uk/publications/bulletins/april_2013.cfm (accessed 17 January 2021).

5.24 Loss of Hermes 450

On 2 October 2011, a Hermes 450 [UAS](#) crashed at Bastion Airfield, Afghanistan. The aircraft was unrepairable.

The aircraft sortie was terminated early due to rising engine temperature. Due to the presence of vehicles and people in the vicinity of the runway, the [GPS take-off and landing system \(GTOLS\)](#) was selected to land the aircraft. Shortly after the approach had been initiated, the landing was self-aborted by the [UAS](#). This abort occurred as a result of an incorrect data parameter in the [GTOLS](#) set-up loaded by the crew.

Moments after the self-abort, due to the urgency of the situation and the additional strain on the engine caused by the aborted landing, the crew chose to abbreviate the pre-programmed go-around [GTOLS](#) route. Instead, they issued a ‘fly to coordinate’ command. As the aircraft was climbing, the engine temperature rose rapidly, before the engine failed completely. On its descent the aircraft initially impacted an unoccupied hangar, before striking the ground upside down. It eventually came to rest on an empty aircraft dispersal pan.

The Service Inquiry determined that the cause of the accident was engine failure, as a result of overheating caused by oil starvation. Like many incidents, there were a considerable number of interacting factors. In total, thirteen contributory factors were identified, including the error in the [GTOLS](#) data.

This incident highlights the importance of the [integrity data property](#), with respect to the [GTOLS](#) data loaded by the crew.

Links

- <https://www.gov.uk/government/publications/service-inquiry-investigating-the-accident-involving-unmanned-air-system-uas-hermes-450-zk515-on-02-oct-11> (accessed 29 November 2017).

5.25 Advocate Lutheran Hospital

A Chicago hospital paid \$8.25 million to settle a lawsuit brought by the parents of an infant boy who died at the institution in October 2010 after a series of medical errors.

The mother gave birth to her son 4 months prematurely. She stayed by his side with her husband for the next six weeks while the boy remained in the hospital’s care. On 15 October, the baby suddenly died after coming out of a heart operation without any clear complications.

The hospital determined that a pharmacy technician had entered [information](#) incorrectly when processing an electronic [intravenous \(IV\)](#) order for the baby. This resulted in an automated machine preparing an [IV](#) solution containing a massive overdose of sodium chloride, more than 60 times the amount ordered. The problem would have been identified by automated alerts in the [IV](#) compounding machine, but these were not activated when the customised bag was prepared for the baby. That is, [adaptation data](#) had been used to change the behaviour of the machine.

Investigations also found that the outermost label on the [IV](#) bag administered to the baby did not reflect its actual contents. Furthermore, although a blood test on the infant had shown abnormally high sodium levels, a lab technician assumed the reading was inaccurate. This highlights a different perspective on the dangers of defaulting, in this case a default assumption rather than a numerical default value.

Since the incident, staff have been activating alerts for similar [IV](#) compounders used in the system’s hospitals and strengthened “double check” policies for all medications leaving pharmacies.

This incident highlights the importance of the [integrity](#) and [verifiability data properties](#), for example with regard to: the [information](#) in the [IV](#) order; the bag label; and the blood test results.

Links

- http://articles.chicagotribune.com/2012-04-05/news/chi-parents-awarded-825-million-in-infants-death-20120405_1_clear-complications-lab-technician-double-check-policies (accessed 29 November 2017).

5.26 Grounding of Sichem Osprey

On 10 February 2010 at 0436 (local), the chemical tanker *Sichem Osprey*, on her way from Panama to Ulsan (South Korea) stranded at more than 16 knots on the north-easterly part of Clipperton Island. An [officer of the watch \(OOW\)](#) and a lookout were on the bridge at the time and no damage had been reported prior to the accident. A 100 metre fore part of the vessel had been grounded. No pollution was observed.

Anti-collision radar alarm thresholds were apparently not set according to the Captain's instructions. There were also sizeable discrepancies between the fixes plotted on the chart and those displayed on the radar.

This incident highlights the role of the [integrity data property](#), with respect to the chart plots, and the [accuracy data property](#), with respect to the alarm thresholds which did not reflect the Captain's wishes.

Links

- <https://www.nautinst.org/download.cfm?docid=F9DA081F-6C1E-40F0-A71F0A89B10F426C> (accessed 5 December 2017).
- http://www.bea-mer.developpement-durable.gouv.fr/IMG/pdf/rET_SICHEM_OSPrEY_05-2010_Site.pdf (in French) (accessed 29 November 2017).

5.27 Near Collision of Trains, Cootamundra

On 12 November 2009, a passenger train was being routed into Number 1 Platform road at Cootamundra, New South Wales. The driver of the passenger train received a signal indicating that the route was clear. However, as he approached, he noticed that the last wagon of a freight train was blocking his path. He applied the train brakes and stopped just short of a collision.

The investigation determined that a signalling system design error had allowed the incorrect signal to occur. The error happened despite the staff involved being suitably qualified and experienced. Working against a tight timescale, they were simultaneously developing a control table and associated software, rather than adopting the normal sequential approach. The control table contains [information](#) on points, signal and level crossing interlocking logic. The tight timescale also compromised the normal testing process. In addition, the quality control process was somewhat lacking: for example, not all identified queries and issues were appropriately closed out.

This incident illustrates the importance of the [integrity data property](#), with respect to the control table data, and the [completeness data property](#), with respect to data produced by the testing and the quality control processes.

Links

- http://www.atsb.gov.au/publications/investigation_reports/2009/rair/ro-2009-009.aspx (accessed 29 November 2017).

5.28 Cedars-Sinai Medical Center Scanner

A software misconfiguration in a [computed tomography \(CT\)](#) scanner used for brain perfusion scanning at Cedar Sinai Medical Center in Los Angeles, California, resulted in 206 patients receiving radiation doses

approximately eight times higher than intended. This error persisted for an 18 month period, starting in February 2008. Some patients reported temporary hair loss and erythema.

The problem reportedly arose from an error made by the hospital in resetting the CT machine after it began using a new protocol for the procedure in February 2008. The error was not detected until one of the patients reported patchy hair loss in August 2009. “There was a misunderstanding about an embedded default setting applied by the machine,” according to a statement from Cedars-Sinai. “As a result, the use of this protocol resulted in a higher than expected amount of radiation.”

This incident highlights the importance of the *verifiability data property*, especially with regards to default (and *adaptation data*).

Links

- <http://articles.latimes.com/2009/oct/10/local/me-cedars-sinai10> (accessed 29 November 2017).

5.29 Grounding of The Pride of Canterbury

On 31 January 2008, the roll-on roll-off passenger ferry, *Pride of Canterbury* grounded on a charted wreck while sheltering from heavy weather in an area known as ‘The Downs’ off Deal, Kent. The vessel suffered severe damage to her port propeller system but was able to proceed unaided to Dover, where she berthed with the assistance of two tugs.

The vessel had been in the area for over 4 hours when, while approaching a turn at the northern extremity, the bridge team became distracted by a fire alarm and a number of telephone calls for *information* of a non-navigational nature. The vessel overshot the northern limit of the identified safe area before the turn was started. The OOW became aware that the vessel was passing close to a charted shoal, but he was unaware that there was a charted wreck on the shoal. The officer was navigating by eye and with reference to an ECS which was sited prominently at the front of the bridge, but he was untrained in the use and limitations of the system. The wreck would not have been displayed on the electronic chart due to the user settings in use at the time. A paper chart was available, but positions had only been plotted on it sporadically and it was not referred to at the crucial time.

Although the *voyage management system (VMS)* was loaded with *electronic navigational charts (ENCs)* for the vessel’s area of operation, the system had not been approved by the *Maritime and Coastguard Agency (MCA)* as the owner’s policy was for the VMS to be used as an aid to navigation only, with *Pride of Canterbury’s* paper charts being used as the primary means for navigation. relevant admiralty charts were supplied to the vessel for this purpose.

Despite the VMS being unapproved for use as the primary means of navigation, the officers on *Pride of Canterbury* were apparently using it as if it was. Furthermore, many of the officers, including the Chief Officer, who was in charge at the time of the accident, were not fully trained in the use of the system.

This incident highlights the importance of the *accuracy data property*, with regards to the *information* displayed on the electronic chart. It also highlights the importance of the *completeness data property*, with regards to training (and training records) and the *intended destination / usage data property*, with regards to the inappropriate use of the VMS data.

Links

- <https://assets.publishing.service.gov.uk/media/547c700ded915d4c0d000071/PrideofCanterburyreport.pdf> (accessed 29 November 2017).

5.30 LOT Flight 282

On 4 June 2007, just after take-off from runway 09r at [LHR](#), the pilots noticed that most of the [information](#) on both of the Electronic Attitude Director Indicators and Electronic Horizontal Situation Indicators had disappeared. The aircraft entered [instrument meteorological conditions \(IMC\)](#) at about 1,500 feet [above aerodrome level \(AAL\)](#), and the co-pilot had no option but to fly using the standby attitude indicator and standby compass. He experienced difficulty in following radar headings. The aircraft returned to land at [LHR](#) after a flight of 27 minutes.

A single error made by the co-pilot during the pre-flight preparation caused the subsequent problems. This was the use of 'E' instead of 'W' when the longitude coordinates were entered into the [FMS](#).

The airports around London, because of their proximity to the Prime Meridian, can lead flight crews to make coordinate entry errors of this nature. It is of note that the operator's route network is such that there are few destinations to the west of the Prime Meridian and hence the majority of longitude coordinates that need to be entered would be eastings. [inertial reference system \(IRS\)](#) alignment warnings should have alerted the crew but may have been dismissed.

This incident highlights the importance of the [integrity](#) and [fidelity / representation data properties](#), specifically with respect to coordinates.

Links

- https://www.gov.uk/government/uploads/system/uploads/attachment_data/file/384859/Bulletin_6-2008.pdf (accessed 29 November 2017).

5.31 Annabella Container Ship – Baltic Sea

On the evening of the 25 June 2007 the container ship the Annabella was subjected to heavy seas causing the vessel to pitch and roll heavily. The following morning the ship's crew discovered that, due to the induced stresses, a "stack of seven 30 ft cargo containers" had collapsed resulting in crushing damage to the lowest containers. A number of these containers were transporting Class 2 Dangerous Goods in the form of Butylene gas.

The Marine Accident Investigation Branch (MAIB) concluded that the container stack had been "piled too high both for the particular hold location and the stacking limits of the containers". The MAIB identified that one of the incident's contributing factors had been an incorrect loading plan which had been produced by planning software used by the cargo company. The software application should have taken account of the stability and stowage [information](#) pertinent to the vessel (as provided by its manufacturer). The application, however, had unknowingly converted the container's dimensions from 30 ft to 40ft resulting in the wrong stacking limits being detailed. The cargo company passed the loading plan to the shipping terminal prior to it being inputted to the vessel's on-board loading computer. The computer did not recognize the error and the 40 ft limits were applied. Amongst the MAIB's conclusions it was noted that,

although the master is responsible for the final loading plan, appropriate oversight is difficult in practice in light of the “pace of modern container operations”. The MAIB made the following recommendations in relation to the [information](#) system:

- Loading computer programs should incorporate the full requirements of a vessel’s cargo securing manual and be properly approved to ensure that officers can place full reliance on the [information](#) provided;
- The [availability](#) of a reliable and approved loading computer programme is a factor to be considered in determining an appropriate level of manning for vessels on intensive schedules;
- Cargo planning software should be able to recognize and alert planners to the consequences of variable data, such as non-standard container specifications.

This incident involved incorrect data, that could have been identified if the software had highlighted the unusual aspects of the data such as the container dimensions to the operator. From a data perspective, the properties that needed to be maintained were [integrity](#), [verifiability](#) and [fidelity / representation](#).

Links

- Marine Accident Investigation Branch report 21 – *report on the Investigation of the Collapse of Cargo Containers Annabella Baltic Sea 26 February 2007* [online] available at <https://assets.publishing.service.gov.uk/media/547c7032e5274a429000007d/Annabellareport.pdf> (accessed 17 January 2021).

5.32 Comair Flight 5191

On 27th August 2006, Comair flight 5191 crashed during take-off from Blue Grass Airport, Lexington, Kentucky. The flight crew was instructed to take off from runway 22, but instead lined up on runway 26 and began the take-off roll. The airplane ran off the end of the runway and impacted the airport perimeter fence, trees, and terrain. The captain, flight attendant and 47 passengers were killed.

The National Transportation Safety Board determined that the probable cause of the accident was the flight crews’ failure to use available cues and aids to identify the airplane’s location on the airport surface during taxi and their failure to cross-check and verify that the airplane was on the correct runway before take-off.

The Airport Charts used by the crew were inaccurate. The airport was under construction, and the charts were not kept current with the rapid changes that were taking place during the construction work. The chart did not accurately reflect either the taxiway identifiers or a taxiway that was closed on the day of the accident.

Due to a previously unrecognized software problem, any [information](#) the chart provider received after normal work hours on Fridays was not included in their regular updates. Furthermore, the chart provider modified the Blue Grass Airport chart after the accident to include a note that runway 8/26 is “daytime [VMC](#)¹ use only”, even though this [information](#) had been published since 2001. Additionally there was a local [NOTAM](#) issued advising of taxiway closures due to construction work. However the crew was not provided with this [information](#) in their dispatch paperwork.

This incident highlights the importance of the [timeliness data property](#), specifically with regards to the charts.

¹ [visual meteorological conditions](#)

The [completeness data property](#) is also relevant, given that neither the “after hours” changes on Fridays nor the local [NOTAM](#) were communicated appropriately.

Links

- <http://libraryonline.erau.edu/online-full-text/ntsb/aircraft-accident-reports/AAr07-05.pdf> (accessed 29 November 2017).
- http://en.wikipedia.org/wiki/Comair_Flight_5191 (accessed 29 November 2017).

5.33 Überlingen Mid-Air Collision

On 1 July 2002, a passenger jet (Bashkirian Airlines) and a cargo jet (DHL Flight 611) collided in mid-air. The collision happened over the south German town of Überlingen; it occurred despite both aircraft being equipped with [TCAS](#).

The two aircraft were in airspace that was controlled from Zürich and were on a collision course. A single controller was on duty and they were responsible for controlling two workstations. This arrangement was against regulations, but was tolerated by management and had become accepted practice. Initially, the controller did not appreciate the dangerous situation that was developing.

Maintenance was being conducted on the main radar system, which meant that the controller was reliant on a backup system. This delayed the presentation of radar [information](#). In addition, a ground-based optical system that would have provided warning of the impending collision was turned off, also for maintenance: the controller was unaware of this.

Less than a minute before the collision the controller became aware of the situation. He instructed Flight 2937 to descend. Seconds after initiating this decent, the [TCAS](#) on Flight 2937 requested a climb, with the corresponding system on Flight 611 requesting a descent. Flight 2937 continued to follow the controller’s direction, meaning that both aircraft were descending.

Unaware of the [TCAS](#) instructions the controller repeated the request for Flight 2937 to descend; he also provided Flight 2937 with misleading [information](#) on the relative location of Flight 611. The planes collided, resulting in the deaths of all 69 people on Flight 2937 and both people on Flight 611.

One of the actions resulting from the accident was a clarification from [ICAO](#) of how pilots should respond to contradictory [information](#) from a controller and [TCAS](#).

This incident illustrates the importance of the following [data properties](#): [consistency](#), with regards to the instructions provided to Flight 2937; [availability](#), with regards to [information](#) from the ground-based optical system; and [timeliness](#), with regards to [information](#) from the radar system.

Links

- https://en.wikipedia.org/wiki/Überlingen_mid-air_collision (accessed 29 November 2017).

5.34 Fort Drum Artillery Incident

Two artillery shells were fired more than a mile off target during an Army firing exercise at Fort Drum in Northern New York in March 2002. The shells landed near a mess tent where a Battalion were having

breakfast. Two soldiers were killed, 13 were injured.

The initial artillery site was unsuitable so the unit had to move to a different location nearly a mile away. The unit then had trouble setting up its digital and wire communications. The movement of the unit was not taken into account when programming the firing coordinates. Also, in what was termed a ‘software behavioural shortfall’ the system was designed to reset the gun elevation to zero. The correct altitude for the new site was not entered into the safety calculations, and the mistakes were not captured by the data review process.

This incident highlights the importance of the *integrity* and *verifiability data properties*, specifically with respect to the location and elevation data.

Sources

- <http://www.apnewsarchive.com/2002/Army-reports-on-Ft-Drum-Accident/id-539bf2ea24b8dd66009c6efee2be926c> (accessed 29 November 2017).

5.35 Early Release from Washington State Prison

For over 13 years the Washington State *Department of Corrections (DoC)* had been releasing certain prison inmates earlier than their sentences allowed.

In 2002 the Supreme Court ruled that the *DoC* was erroneously denying offenders credit for early release time earned during pre-sentence detention. In attempting to address that issue the *DoC* incorrectly reprogrammed its computer tracking system. This resulted in the early release of offenders with sentencing enhancements. The programming error went undetected for over ten years, with more than 2,000 offenders being released early.

The error was detected when the family of an assault victim hand-calculated the assailant’s release date. The family notified the *DoC* that it appeared as if the assailant would be released earlier than warranted by statute. It took a further three years before the programming error was finally corrected.

This incident illustrates the importance of the *integrity data property*, with respect to calculated release dates. The *verifiability data property* is also relevant, noting that the calculation of the release date was readily verifiable (as shown by the actions of the family of the assault victim).

Links

- http://www.governor.wa.gov/sites/default/files/documents/2016-02-25_DOC_report.pdf (accessed 29 November 2017).

5.36 Mars Climate Orbiter

The Mars Climate Orbiter was a spacecraft launched aboard a Delta II rocket by NASA from Cape Canaveral on 11th December 1998. Its intended mission was to study the Martian atmosphere and climate, while acting as a communications relay for other spacecraft on or near Mars.

The plan was that the rocket would place the spacecraft into a transfer orbit to Mars, which would be optimized along the way by a series of four trajectory correction manoeuvres. Insertion into Mars orbit was

to take place at an altitude of 226 km, but during the week after the final correction manoeuvre, calculations predicted that it would be between 150 km and 170 km; revised to 110 km the day before insertion. The orbiter was able to survive atmospheric stresses down to about 80 km. On 23rd December 1999, the spacecraft passed behind Mars, and so out of radio contact, earlier than expected; communications were never regained.

Final calculations placed the spacecraft in a trajectory that would have taken it within 57 km of the Martian surface, but it is likely to have disintegrated before getting to that point.

It transpires that the orbiter's [FMS](#) software was designed to work with metric Newton seconds, whereas a [FMS](#) data file generated by ground system software used pound-force seconds. A Newton is about 22.5% of a pound-force or a factor of 4.45.

The cost of the mission was stated by NASA to have been \$327.6 million in total (\$193.1 million to develop the spacecraft, \$91.7 million for launch and \$42.8 million for mission operations).

This incident highlights the importance of the [consistency data property](#).

Links

- http://en.wikipedia.org/wiki/Mars_Climate_Orbiter (accessed 29 November 2017).

5.37 Crash into Nimitz Hill, Guam

On 6 August 1997, Korean Air Flight 801 crashed at Nimitz Hill, Guam. This is high terrain approximately 3 miles southwest of Guam International Airport, where the aircraft had been cleared to land. Of the 254 people on board, 228 were killed and 26 survived with serious injuries.

Probable causes of the accident were the captain's failure to adequately brief and execute a non-precision approach and the first officer's and flight engineer's failure to effectively monitor and cross-check this approach. Contributing factors included fatigue and inadequate flight crew training.

Another contributing factor was the intentional inhibition by the [Federal Aviation Administration \(FAA\)](#) of the [MSAW](#) system at Guam, and the agency's failure to adequately manage the system.

The [MSAW](#) system uses a terrain [database](#). It is designed to alert a controller if an aircraft equipped with a Mode C transponder descends below, or is predicted to descend below, a predetermined safe altitude.

In 1990, the Guam terminal [MSAW](#) was installed to provide protection within a 55 nm radius. In 1993, a new software package was produced in which warnings were inhibited within a 54 nm radius; this left a 1 nm annular region within which warnings could be generated. The motivation behind the new configuration was to reduce false alarms. The software became operational in February 1995. A further software update became operational in April 1996. This also had the 54 nm inhibition.

This incident illustrates the importance of the [continuity data property](#), with respect to the [MSAW](#) coverage, and the [fidelity / representation data property](#), regarding the terrain [database](#) used by the [MSAW](#) system.

Links

- <https://www.nts.gov/investigations/Accidentreports/reports/AAR0001.pdf> (accessed 29 November 2017).

5.38 San Bernardino Derailment and Pipeline Rupture

In May 1989 a Southern Pacific Transportation Company freight train derailed in San Bernardino, California. The train derailment accounted for seven fatalities and two serious injuries; however, that accident also damaged a fuel pipe and less than a fortnight later it ruptured causing a further two deaths and three serious injuries.

One of the causal factors of the train's derailment, as reported by the National Transportation Safety Board, was a "failure to determine the weight of the train" and in summary, the operator thought it weighed less than it actually did, resulting in the dynamic braking being insufficient to deal with the downhill gradient it was travelling on. The Company had used a computer to determine the train's weight and because the actual weights had not been entered the system made its calculations based upon estimated weights, which were lower. Clearly this was not a systematic failure of the computation algorithm but again potentially a failure to appreciate the [criticality](#) of the weight [information](#) and its potential as a causal factor within an accident sequence.

From the [criticality](#) of the weight [information](#), derived safety requirements could be developed for the various data sources that were used to derive the weight. Such requirements could be expected to highlight the properties of [integrity](#), [completeness](#), [accuracy](#), [timeliness](#), [verifiability](#), [fidelity / representation](#) and [lifetime](#).

T Hardy, *Software and System Safety – Accidents, Incidents and Lessons Learned*, Bloomington, Author House, 2012, ISBN 978-1-4685-7470-8

5.39 Lake Peigneur Drilling Accident

Lake Peigneur is located in Louisiana, United States of America. It was a ten-foot deep freshwater lake popular with sportsmen. On 20th November 1980, an exploration rig drilling for oil in the lake bed was evacuated as it began to sink; this was perceived by the crew as a structural collapse. Meanwhile, the nearby Jefferson Island salt mine was being evacuated due to the sudden onset of flooding.

The rig crew had been drilling a test well into deposits alongside a salt dome under Lake Peigneur. By some miscalculation, the assembly drilled into the third level of the nearby salt mine. Fresh water from the lake soon began trickling into the mine. Over the course of the morning, the fresh lake water began dissolving the salt and enlarging the hole until water was literally flooding into the mine.

The whirlpool created as the lake drained into the mine sucked in the drilling platform, eleven barges, trees and soil. The Delcambre Canal, which usually drains from the lake into a bay on the Gulf of Mexico, had its flow reversed. This resulted in Lake Peigneur becoming a salt water lake. Fortunately, no injuries or loss of human life were reported.

Federal experts from the Mine Safety and Health Administration were not able to determine the cause of the accident due to confusion over whether the rig was drilling in the wrong place or whether the mine's maps were inaccurate.

This incident highlights the importance of the [verifiability data property](#), specifically with regards to the location of the rig. Note that this property was relevant both when the rig started to drill and also during the post-incident investigation.

Links

- http://en.wikipedia.org/wiki/Lake_Peigneur (accessed 29 November 2017).

This page is intentionally blank

6 Acronyms, Definitions and Glossary (Discursive)

Most controversies would soon be ended, if those engaged in them would first accurately define their terms, and then adhere to their definitions

Tryon Edwards

6.1 Acronyms

AAL	above aerodrome level
AI	artificial intelligence
AoA	angle of attack
ATSB	Australian Transport Safety Bureau
CCRC	Criminal Cases Review Commission
CFIT	controlled flight into terrain
COTS	commercial off-the-shelf
CSV	comma separated variable
CT	computed tomography
DME	distance measuring equipment
DoC	Department of Corrections
DSAL	data safety assurance level
DSG	data safety guidance
DSIWG	Data Safety Initiative Working Group
ECS	electronic chart system
ECU	electronic control unit
EDM	entry demonstrator module
EGPWS	enhanced ground proximity warning system
EHR	electronic health record
ENC	electronic navigational chart
ESA	European Space Agency
FAA	Federal Aviation Administration

FMGS	flight management guidance system
FMS	flight management system
GCS	ground control station
GTOLS	GPS take-off and landing system
HAZOP	hazard and operability study
HUMS	health and usage monitoring system
IBM	International Business Machines Corporation
ICAO	International Civil Aviation Organization
IMC	instrument meteorological conditions
IMU	inertial measurement unit
IP	internet protocol
IRS	inertial reference system
IV	intravenous
LHR	London Heathrow airport
LLM	large language model
MCA	Maritime and Coastguard Agency
MCAS	manoeuvring characteristics augmentation system
ML	machine learning
MSAW	minimum safe altitude warning
NOTAM	notice to airmen
OED	Oxford English Dictionary
OOW	officer of the watch
PCR	polymerase chain reaction
POCL	Post Office Counters Limited
RDA	radar Doppler altimeter
SAR	search and rescue
SCSC	Safety-Critical Systems Club
SPM	subpostmaster
SSS	Safety-critical Systems Symposium

TCAS	traffic collision avoidance system
UAS	unmanned air system
VHF	Very High Frequency
VMC	visual meteorological conditions
VMS	voyage management system
VOR	VHF omnidirectional range

6.2 Definitions and Glossary

accessibility

Property that the data is visible only to those that should see it.

accuracy

Closeness of agreement between a measured quantity value and a true quantity value of a measurand. [15]

adaptation data

Data used to customise elements of the system for their designated purpose. Adaptation data is used to customise elements of the system for its designated purpose at a specific location. These systems are often configured to accommodate site-specific characteristics. These site dependencies are developed into sets of adaptation data. Adaptation data includes data that configures the software for a given geographical site, and data that configures a workstation to the preferences and / or functions of an operator. Examples include, but are not limited to:

Geographical Data: latitude and longitude of a radar site.

Environmental Data: operator selectable data to provide their specific preferences.

Airspace Data: sector-specific data.

Procedures: operational customisation to provide the desired operational role.

Adaptation data may take the form of changes to either database parameters or take the form of pre-programmed options. In some cases, adaptation data involves re-linking software code to include different libraries. Note that this should not be confused with recompilation in which a completely new version of the code is generated. Based on ED-153 [16]

aeronautical data

- A representation of aeronautical facts, concepts or instructions in a formalized manner suitable for communication, interpretation or processing. (EU) No 73/2010 [17]
- Data used for aeronautical applications such as navigation, flight planning, flight simulators, terrain awareness, and other purposes. RTCA/DO-178C [18]

aliasing

An effect that causes different data to become indistinguishable when accessed

analysability

The data (including any [metadata](#)) is of a suitable size, type and format to enable it be usefully analysed

artefact, data

An item, or collection of items, that provides a useful perspective on data generated, processed or consumed by a system.

assurance level, software

An indication of how much assurance is required (commensurate to risk) before deploying software into an operational system. J Spriggs, based on (EC) No 482/2008 [19]

availability

The property of being accessible and usable upon demand by an authorized entity. ISO 27001:2013 [20]

clarity

The property that The data is visible, unmasked, not hidden or obscured, and not camouflaged

completeness

- Property of having every necessary part or element.
- Completeness of the data provided. RTCA/DO-200A

confidentiality

The property that [information](#) is not made available or disclosed to unauthorized individuals, entities, or processes. ISO27001:2013 [20]

configuration data

- Data that configures a generic software system to a particular instance of its use. (EC) No 482/2008 [19]
- Data that configures a generic software system to a particular instance of its use (e.g., data for flight data processing system for a particular airspace, by setting the positions of airways, reporting points, navigation aids, airports and other elements important to air navigation). ED-153 [16] Data that configures a generic software system to a particular instance of its use (e.g., data for flight data processing system for a particular airspace, by setting the positions of airways, reporting points, navigation aids, airports and other elements important to air navigation). ED-153 [16]

consistency, data

The property that the data adheres to a common world view (e.g., units).

continuity, data

The property that the data is continuous and regular without gaps or breaks.

correctness, data

self-[consistency](#), protection against alteration or corruption and [consistency](#) with the functional requirements of the [data-driven system](#). IEC 61508 Part 3 [21]

criticality, data

Classification of data by the potential effect of erroneous data on the expected operation that is supported by that data. RTCA/DO-200A [22]

data

- A thing given or granted; something known or assumed as fact, and made the basis of reasoning or calculation; an assumption or premiss from which inferences are drawn. [Oxford English Dictionary \(OED\)](#)
- A reinterpretable representation of [information](#) in a formalized manner suitable for communication, interpretation or processing. ISO/IEC 2382 [23]

database

A set of data, part or the whole of another set of data, consisting of at least one file that is sufficient for a given purpose or for a given data processing system. RTCA/DO-178C

data-driven system

System which relies upon [configuration data](#) or lookup tables to define the functionality of the system. IEC 61508 Part 4 [24]

dataset

Identifiable collection of data. Note that a dataset may be a smaller grouping of data which, though limited by some constraint such as spatial extent or feature type, is located physically within a larger dataset. Theoretically, a dataset may be as small as a single feature or feature attribute contained within a larger dataset. A hardcopy map or chart may be considered a dataset. BS EN ISO 19131:2008 [25]

disposability / deletability

The property that the data can be permanently removed when required

error, data

- Discrepancy with the universe of discourse. ISO 19138:2006 [26]
- Discrepancy between a data value and the true, specified or theoretically correct value or condition. P. Ensor [27]

explainability

The property that the data can be meaningfully explained, by a suitable mechanism, to those who need to understand it.

fidelity / representation ,data

The property describing how well the data maps to the real world entity it is trying to model.

format

The property that data is represented in a which is readable by those that need to use it.

Goldilocks

The property that the data is just the right size – not too much and not too little

hazard log

A record of all hazard analysis, safety risk assessment and safety risk reduction activities for the “whole-of-life” of a safety-related system.

hazard, data

Use of data (in the context of a system) that could lead to harm. [SCSC DSIWG](#)

history

Property that the data has an audit trail of changes.

information

- Knowledge communicated concerning some particular fact, subject, or event; that of which one is apprised or told - intelligence, news - as contrasted with data. [OED](#)
- Knowledge that has a contextual meaning. ISO/IEC 2382 [23]

information, aeronautical

[Information](#) resulting from the assembly, analysis and formatting of [aeronautical data](#). (EU) No 73/2010 [17]

integrity, data

- The assurance that a data element retrieved from a storage system has not been corrupted or altered in any ways since the original data entry or latest authorised amendment. RTCA/DO-200A [22]
- The degree of assurance that a [data item](#) and its value have not been lost or altered since the data origination or authorised amendment. (EU) No 73/2010 [17]
- The degree of undetected (at system level) non-conformity of the input value of the [data item](#) with its output value. (EU) No 1207/2011 [28]
- The property of protecting the [accuracy](#) and [completeness](#) of assets, i.e., that which has value to the organization. ISO 27001:2013 [20]

intended destination / usage

Property that the data is only sent to those that should have access to it

item, data

Single attribute of a complete [dataset](#), which is allocated a value that defines its current status. (EU) No 73/2010 [17]

lifetime

The property of when the safety-related data expire

metadata

Data that represents [information](#) about data itself. Note that one should distinguish between “Structural Metadata”, which is data about the design and specification of data structures (and is more properly called “data about the containers of data”) and “Descriptive Metadata”, which is about individual instances of application data, the data content. J. Inge [3]

mitigation

Steps taken to control or prevent a hazard from causing harm and reduce risk to a tolerable or acceptable level. [29]

owner, data

The individual or organization responsible for a particular [data artefact](#) or collection of [data artefacts](#).

priority

The property that data is presented / transmitted / made available in the order required.

property, data

A characteristic that can be exhibited by a [data artefact](#).

quality, data

- A degree or level of confidence that the data provided meet the requirements of the user. These requirements include levels of [accuracy](#), [resolution](#), [DSAL](#), [traceability](#), timeliness, [completeness](#), and format. RTCA/DO-200A [22]
- A degree or level of confidence that the data provided meets the requirements of the data user in terms of [accuracy](#), [resolution](#) and [integrity](#). (EU) No 73/2010 [17]

resolution

- The ability of a device to respond to small differences in input and to indicate or represent them accurately in output; a measure of this, expressed as the smallest difference so distinguishable. [OED](#).
- The smallest difference between two adjacent values that can be represented in a data storage, display or transfer system. RTCA/DO-200A [22]
- The number of units or digits to which a measured or calculated value is expressed and used. (EU) No 73/2010 [17]

response

The way in which an identified risk is addressed; possible responses include avoid / eliminate, treat, or accept as sufficiently low.

safety assessment, data

The process of explicitly considering data as part of a system safety assessment, via the means of [data artefacts](#), Data Properties and [DSALs](#).

safety assurance level, data

An indication of the level of rigour with which relevant [data properties](#) should be demonstrated for appropriate [data artefacts](#).

safety requirement, data

A requirement to implement an approach specifically designed to achieve, maintain or demonstrate a [data property](#) (or [data properties](#)) for a given [data artefacts](#) (or Artefacts).

sequencing

The property that the data is preserved in the order required.

stakeholder

An individual or organization that has some relationship to the system, possibly including a power of veto.

suppression

Property that the data is intended never to be used again

tailoring

Adaptation of processes etc. to be appropriate for a specific system and context

timeliness

- A measure of the time delay between a change in the real world and the associated [database](#) update being available to the user. P. Ensor [27]
- The difference between the time of output of a [data item](#) and the time of applicability of that [data item](#). (EU) No 1207/2011 [28]

traceability

Ability to determine the origin of the data. RTCA/DO-200A [22]

treat

To apply a [treatment](#)

treatment

An action taken to reduce or control risk.

validation, data

- The activity whereby a data element is checked as having a value that is fully applicable to the identity given to the data element, or a set of data elements that is checked as being acceptable for their purpose. RTCA/DO-200A [22]
- Process of ensuring that data meets the requirements for the specified application or intended use. (EU) No 73/2010 [17]

verifiability

Evaluation of the output of an [aeronautical data](#) process to ensure [correctness](#) and [consistency](#) with respect to the inputs and applicable data standards, rules and conventions used in that process. (EU) No 73/2010 [17]

The normative list of definitions is at **??**. Normative definitions have been repeated here for convenience.

7 References

Data opportunities multiply as the data is transformed.
Sun Tzu misquoted

Bibliography

- [1] Vocabulary.com. homonym vs. homophone vs. homograph : Choose your words. URL <https://www.vocabulary.com/articles/chooseyourwords/homonym-homophone-homograph/>. Accessed: 17 January 2023.
- [2] Alastair Faulkner and Mark Nicholson. *Data-Centric Safety: Challenges, Approaches, and Incident Investigation*. Elsevier, June 2020. ISBN 978-0128207901. Available from <https://www.amazon.co.uk/dp/0128207906?ie=UTF8&n=341677031>. Accessed 30 January 2021.
- [3] J Inge. *Improving the Analysis of Data in Safety-Related Systems*. University of York, September 2008.
- [4] Software Tool Qualification Considerations. Standard RTCA/DO-330, EUROCAE/ED-215, Radio Technical Commission for Aeronautics / European Organisation for Civil Aviation Equipment, January 2012.
- [5] Die Lage der IT-Sicherheit in Deutschland 2014. URL <https://www.bsi.bund.de/SharedDocs/Downloads/DE/BSI/Publikationen/Lageberichte/Lagebericht2014.pdf>, December 2014. Accessed: 27 January 2021.
- [6] D LeBlanc and M Howard. *Writing secure code*. Microsoft Press, December 2002.
- [7] *Cyber Security and Safety Code of Practice*. IET, 2020. ISBN 978-1-83953-318-8.
- [8]
- [9] Barack Obama. One election won't stop us 'truth decay'. URL <https://www.bbc.co.uk/news/election-us-2020-54910344>.
- [10] *Systems and Covid-19, Proceedings of the 29th Safety-Critical Systems Symposium (SSS'21)*, SCSC-161. SCSC, 2021. ISBN 979-8588665049. Available from <https://www.amazon.co.uk/dp/B08RRMSBTN>. Accessed 30 January 2021.
- [11] The Honourable Mr. Justice Frazer. Alan bates and others -v- post office limited: Technical appendix to judgement (no.6) "horizon issues". URL <https://www.judiciary.uk/wp-content/uploads/2019/12/bates-v-post-office-appendix-1.pdf>, 2019. Accessed: 30 January 2024.
- [12] Karl Flinders. Post office horizon scandal explained: Everything you need to know. *Computer Weekly*, 2024. Accessed: 30 January 2024.
- [13] The ccrc and post office/horizon cases. URL <https://ccrc.gov.uk/news/the-ccrc-and-post-office-horizon-cases/>, 2024. Accessed: 30 January 2024.
- [14] The Honourable Mr. Justice Frazer. Alan bates and others -v- post office limited: Judgement (no.6) "horizon issues". URL <https://www.judiciary.uk/wp-content/uploads/2019/12/bates-v-post-office-judgment.pdf>, 2019. Accessed: 30 January 2024.
- [15] ISO/IEC. Information technology — data governance — data collaboration framework. <https://www.iso.org/obp/ui#iso:std:iso-iec:25642:ed-1:v1:en:term:3.9>. Accessed: 2 October 2025.

- [16] Guidelines for ANS software safety assurance. Standard EUROCAE/ED-153, European Organisation for Civil Aviation Equipment, August 2009. Used for definitions only.
- [17] Commission Regulation (EU) No 73/2010 of 26 January 2010 laying down requirements on the quality of aeronautical data and aeronautical information for the single European sky. URL <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2010:023:0006:0027:EN:PDF>, January 2010. Accessed: 27 January 2021.
- [18] Software Considerations in Airborne Systems and Equipment Certification. Standard RTCA/DO-178C, EUROCAE/ED-12C, Radio Technical Commission for Aeronautics / European Organisation for Civil Aviation Equipment, January 2012.
- [19] Commission Regulation (EC) No 482/2008 of 30 May 2008 establishing a software safety assurance system, as amended. URL <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32008R0482&qid=1611781633022>, May 2008. Accessed: 27 January 2021, but no longer in force.
- [20] Information Technology — Security Techniques — Information Security Management Systems — Requirements. Standard BS ISO/IEC 27001:2013, International Standards Organisation, 2013.
- [21] Functional safety of electrical/electronic/ programmable electronic safety-related systems: Software requirements. Standard BS EN 61508-3:2010, International Electrotechnical Commission, June 2010.
- [22] Standards for processing aeronautical data. Standard RTCA/DO-200A, EUROCAE/ED-76, Radio Technical Commission for Aeronautics / European Organisation for Civil Aviation Equipment, September 1998.
- [23] Information Technology. Vocabulary. Part 1: Fundamental terms. Standard ISO/IEC 2382-1:1993, International Standards Organisation, 1993.
- [24] Functional safety of electrical/electronic/ programmable electronic safety related systems: Definitions and abbreviations. Standard BS EN 61508-4:2010, International Electrotechnical Commission, June 2010.
- [25] Geographic Information. Data Product Specifications. Standard BS EN ISO 19131:2008, International Standards Organisation, 2008.
- [26] Geographic Information. Data Quality Measures. Standard ISO/TS 19138:2006, International Standards Organisation, 2006.
- [27] P Ensor. *Safety Analysis of Navigational Data*. University of York, September 2009.
- [28] Commission Implementing Regulation (EU) No 1207/2011 of 22 November 2011 laying down requirements for the performance and the interoperability of surveillance for the single European sky. URL <http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2011:305:0035:0052:EN:PDF>, November 2011. Accessed: 27 January 2021.
- [29] Use of safety management systems by ATM service providers. Standard ESARR3, EUROCONTROL, 2011.

8 Acknowledgements

Our ability to do great things with data will make a real difference in every aspect of our lives.
Jennifer Pahlka

The document contributors would like to thank:

- The [SCSC](#).
- The [SCSC](#) Covid-19 Working Group for providing some of the data used in the Covid-19 Appendix.
- Brian Jepson of the [SCSC](#) for web hosting support and technical help with the [SCSC](#) web site.
- Tim Rowe for editing this edition.
- Paul Hampton and Mark Templeton for managing the publication processes.
- Nick Hales, Mike Parsons, Tim Rowe, Alan Simpson and Mark Templeton for developing the additional text for this edition.
- Martin Atkins and Divya Atkins for driving the development of tooling and promoting data safety.
- Mike Parsons for chairing the Working Group meetings.
- All those who have taken minutes at Working Group meetings.
- All the organisations that have hosted Working Group meetings.
- All the organisations that have provided support to the document's contributors.
- Those that have been unable to attend meetings but have made supporting contributions.

This page is intentionally blank

9 Contributors

Without data, you're just another person with an opinion.
W. Edwards Deming

This document has had the benefit of contributions from a large number of people, who work for a variety of organisations, which collectively span a range of different sectors. Note that contributions have been made on an individual basis and, in particular, the inclusion of an organisation in the following list does **not** necessarily mean that organisation agrees with the entire contents of the document.

Updates to the most recent version of the document were written by:

- Divya Atkins, Mission Critical Applications
- Martin Atkins, Mission Critical Applications
- Paul Hampton, CGI IT UK Ltd
- Mike Parsons, Ebeni and [SCSC](#)
- Tim Rowe, TGR Safety Management Ltd

In addition to the above, contributors to earlier versions upon which this document is based include the following (the organisations listed were correct at the time of their contribution) :

- Mike Ainsworth, Ricardo
- Rob Ashmore, Dstl
- Michael Aspaturian, EDF Energy
- Janette Baldwin, Thales UK
- Dave Banham, Blackberry QNX
- Ian Bingham
- John Bragg, MBDA UK Ltd
- Jennifer Brain, Wood plc
- Eric Bridgstock
- Simon Brown, QinetiQ
- Dermot Martin Burke, BAE Systems
- Dale Callicott, DKCSC Ltd
- John Carter, General Dynamics
- Martyn Clarke, SCSS Ltd
- Steve Clugston, TSC
- Robin Cook, Thales
- Davin Crowley-Sweet, Highways England

- Dijesh Das, AMEC / BAE Systems
- Duncan Dowling, DARD
- Andrew Eaton
- Ashraf El-Shanawany, CRA Risk Analysis
- Paul Ensor, Boeing
- Alastair Faulkner, Abbeymeade
- Ken Frazer, KAF
- Richard Garrett, SQEP
- Paulo Giuliani
- Ian Glazebrook, Atkins
- Rob Green, NATS
- Nick Hales
- Louise Harney, Leonardo
- Ali Hessami, Vega Systems
- David Higgins
- Gordon Hurwitz, Thales
- Pete Hutchison, RPS
- Gavin Jones
- Amira Kavar, Kavar Engineering Consultancy Ltd
- Tim Kelly
- Andrew Kent
- Brent Kimberley, Durham, Canada
- Julian Lockett, Frazer-Nash Consultancy Ltd
- David Lund, David Lund Consultants
- Dave Lunn, Thales UK
- Nasser Al Malki, University of York
- Victor Malysz, Rolls-Royce
- Jim Mateer, SQEP
- John McDermid, University of York
- Paul McKernan, Dstl
- Thor Myklebust, Sintef
- Mark Nicholson, University of York

- Yvonne Oakshott
- Robert Oates
- David Perrin, Virtual PV
- Ashley Price, Raytheon UK
- Andrew Rankine
- Felix Redmill, [SCSC](#)
- Sam Robinson, EDF Energy
- Mark Simmonite, Highways England
- Alan Simpson, Ebeni
- Oscar Slotosch, Validas AG
- Dave Smith, Frazer-Nash Consultancy Ltd
- Peter Smith, Highways England
- John Spriggs, NATS
- Carolyn Stockton, BAE Systems
- Mark Templeton, Arcade Experts Ltd
- Andy Williams
- Lesley Winsborrow
- Fan Ye, ESC

This page is intentionally blank

Index of Locations

- Afghanistan
 - Bastion Airfield, [45](#)
- Australia
 - ACT
 - Canberra, [43](#)
 - New South Wales
 - Cootamundra, [28](#), [47](#)
 - Sydney, [42](#)
 - Northern Territory
 - Darwin, [42](#)
 - Western Australia
 - Perth, [43](#)
- Baltic Sea, [49](#)
- France, [31](#)
 - Chambery, [45](#)
 - Clipperton Island, [47](#)
- Guam, [53](#)
 - International Airport, [53](#)
 - Nimitz Hill, [53](#)
- Gulf of Mexico, [54](#)
- Ireland, Republic of
 - Black rock, [38](#)
 - Blacksod, [38](#)
 - Dublin, [38](#)
- Mars
 - Climate Orbiter, [52](#)
 - Lander, Schiaparelli, see Schiaparelli Mars
 - Lander
 - reconnaissance Orbiter, [39](#)
- Nepal, [41](#)
 - Kathmandu, [41](#)
 - Kathmandu-Tribhuvan Airport, [41](#)
- North Sea
 - Haisborough Sand, [43](#)
- Panama, [47](#)
- South Korea
 - Ulsan, [47](#)
- Switzerland
 - Überlingen, [51](#)
 - Zürich, [51](#)
- Türkiye
 - Istanbul
 - Istanbul-Atatürk International Airport, [41](#)
- United Kingdom, [25](#), [29](#), [37](#)
 - England
 - Deal, [48](#)
 - Dover, [48](#)
 - London, [49](#)
 - London Heathrow Airport, [49](#)
 - Shrewsbury, [37](#)
 - Northern Ireland
 - Ardglass, [40](#)
 - Wales
 - Machynlleth, [37](#)
- USA
 - California
 - Los Angeles, [47](#)
 - San Bernardino, [29](#), [54](#)
 - Illinois
 - Chicago, [46](#)
 - Kentucky
 - Lexington, [50](#)
 - Louisiana
 - Delcambre Canal, [54](#)
 - Lake Peigneur, [54](#)
 - Nevada
 - Mount Irish, [44](#)
 - New York
 - Fort Drum, [51](#)
 - New York, [51](#)
 - Texas
 - Dallas, [42](#)
 - Washington State, [52](#)

This page is intentionally blank

Index

- Accessibility Property, [4](#), [8](#), [9](#)
- Accuracy Property, [6–9](#), [14](#), [25](#), [26](#), [28](#), [29](#), [31](#), [32](#),
[34](#), [35](#), [38](#), [41](#), [42](#), [47](#), [48](#), [54](#), [62](#), [63](#)
- Adaptation Data, [36](#), [37](#), [46](#), [48](#), [59](#)
- Advocate Lutheran Hospital, [46](#)
- Airbus
 - A330-303, [41](#)
 - A400M, [40](#)
- Aliasing Property, [25](#), [33](#)
- Analysability Property, [7–9](#)
- Annabella, [49](#)
- Apache, [25](#), [33](#), [34](#)
- Artefact, Data, [3](#), [62](#), [63](#)
- Asset Data, [36](#)
- Assurance Level, [4](#), [63](#)
 - Data, [15](#)
 - Software, [15](#)
- Availability Property, [8](#), [9](#), [25](#), [26](#), [29](#), [35–37](#), [50](#),
[51](#)
- Bashkirian Airlines Flight 2937, [51](#)
- Boeing 737, [27](#), [35](#), [42](#), [43](#), [45](#)
- Butylene, [49](#)
- Cambrian Line, [37](#)
- Category
 - Data, [6](#), [10](#)
- Cedars-Sinai Medical Center, [vi](#), [47](#)
- CFIT, [38](#)
- Clarity Property, [8](#), [9](#)
- Comair Flight 5191, [50](#)
- Completeness
 - Checks, [4](#)
 - Property, [6–9](#), [25–29](#), [35](#), [36](#), [38](#), [40](#), [42](#), [43](#),
[47](#), [48](#), [51](#), [54](#), [62](#), [63](#)
- Completeness Property, [25](#), [31](#)
- Consistency
 - Property, [7–9](#), [27](#), [29](#), [44](#), [51](#), [53](#)
 - Property Handling, [19](#)
 - Self, [60](#)
 - With Inputs, [64](#)
 - With Requirements, [60](#)
- Consistency Property, [25](#), [33](#)
- Continuity Property, [7–9](#), [29](#), [53](#)
- Controlled Flight into Terrain, *see* CFIT
- COTS, [35](#)
- Covid-19, [4](#), [19](#), [25](#), [34](#)
- Dallas Hospital, [42](#)
- Dark Data, [6](#), [35](#)
 - Data We Don't Know are Missing, [35](#)
 - Missing What Matters, [35](#)
- Data
 - Configuration, [47](#)
 - Entry, [27](#), [42](#), [43](#), [62](#)
 - Owner, [9](#)
- Data Property, [4](#), [6–8](#)
 - Accessibility, *see* Accessibility Property
 - Accuracy, *see* Accuracy Property
 - Analysability, *see* Analysability Property
 - Availability, *see* Availability Property
 - Clarity, *see* Clarity Property
 - Completeness, *see* Completeness, Property
 - Consistency, *see* Consistency Property
 - Continuity, *see* Continuity Property
 - Disposability, *see* Disposability Property
 - Explainability, *see* Explainability Property
 - Fidelity, *see* Fidelity Property
 - Format, *see* Format Property
 - Goldilocks, *see* Goldilocks Property
 - History, *see* History Property
 - Integrity, *see* Integrity, Property
 - Intended Destination, *see* Intended
Destination Property
 - Lifetime, *see* Lifetime Property
 - Priority, *see* Priority Property
 - Resolution, *see* Resolution Property
 - Sequencing, *see* Sequencing Property
 - Suppression, *see* Suppression Property
 - Timeliness, *see* Timeliness Property
 - Traceability, *see* Traceability Property
 - Verifiability, *see* Verifiability Property
- DHL Flight 611, [51](#)
- Disposability Property, [8](#), [9](#)
- Dynamic Data, [35](#), [36](#)
- Ebola, [42](#)
- EHR, [42](#)
- Electronic Health record, *see* EHR
- ESA, [38](#)
- Escape sequences, [34](#)
- Ethiopian Airlines, [35](#)
- European Space Agency, *see* ESA
- Evolution
 - System, [14](#)

- Excel, [19](#), [25](#), [34](#)
- Explainability Property, [7–9](#)
- Fidelity / Representation Property, [25–29](#), [31](#), [32](#),
[35–39](#), [43](#), [49](#), [50](#), [53](#), [54](#)
- Fidelity Property, [8](#), [9](#)
- Format Property, [7–9](#), [27](#), [42](#)
- Gemini V, [25](#)
- Goldilocks Property, [6–9](#)
- Gurdasani, Professor Deepti, [34](#)
- HAZOP, [7](#)
- Hermes 450, [45](#)
- History Property, [8](#), [9](#), [25](#), [26](#), [37](#)
- HTML, [34](#)
- Images, Indecent, [39](#)
- Immensa, [34](#)
- Integrity
 - Property, [4](#), [7–9](#), [63](#)
- Integrity Property, [3](#), [15](#), [16](#), [25–29](#), [31](#), [33–37](#),
[39](#), [41](#), [43](#), [46](#), [47](#), [49](#), [50](#), [52](#), [54](#)
- Intended Destination Property, [8](#), [9](#)
- Interface
 - Assessment, [14](#)
- Investigation
 - Criminal, [26](#), [39](#)
 - Incident / Accident, [26](#), [37–39](#), [44–47](#), [49](#), [54](#)
- IST, *see* Istanbul-Atatürk in location index
- Java, [25](#), [33](#)
- Justification Data, [36](#)
- Karen, *see* Trawler Karen
- Known Unknowns, [20](#)
- Korean Air Flight 801, [53](#)
- KTM, *see* Türkiye, Kathmandu-Tribhuvan in location index
- LHR, *see* United Kingdom, England, London
Heathrow Airport in location index
- Lifecycle, [3](#)
 - System, [4](#)
- Lifetime Property, [4](#), [8](#), [9](#), [27](#), [29](#), [45](#), [54](#)
- Lion Air, [35](#)
- Log4j, [25](#), [33](#), [34](#)
- LOT Flight 282, [49](#)
- Machine Learning Data, [6](#)
- Manoeuvring Characteristics Augmentation
System, *see* MCAS
- MCAS, [35](#), [36](#)
- Meteor-M, [26](#), [36](#)
- Minecraft, [33](#)
- Mitigation, [5](#), [14–16](#), [35](#)
- MQ-9 Reaper, [44](#)
- NASA, [39](#), [52](#), [53](#)
- Navigator Scorpio, [43](#), [44](#)
- PCR, [25](#), [34](#)
- Performance Data, [36](#), [42](#), [43](#)
- Polymerase Chain Reaction, *see* PCR
- Predictive Models, [19](#)
- Pride of Canterbury, [48](#)
- Priority Property, [8](#), [9](#)
- Property
 - Data, [i](#), [9](#), [13–16](#), [25–29](#), [35–54](#)
 - Government, [44](#)
 - Safety, [13](#)
 - Statistical, [13](#), [14](#)
 - System Development, [15](#)
- Qantas, [42](#), [43](#)
- Queen Mary University, [34](#)
- Radish, [17](#)
- Radish context, [18](#)
- Railway Signalling, *see* Signalling, Railway
- Resolution Property, [7–9](#), [63](#)
- Rumsfeld, Donald, [20](#)
- Safety Assessment
 - Data, [17](#)
- Safety Requirement, [15](#)
 - Data
 - Derived, [54](#)
- Safety requirement, [45](#)
- Schiaparelli Mars Lander, [38](#)
- Search and Rescue, [38](#)
- Sequencing Property, [8](#), [9](#)
- Sichem Osprey, [47](#)
- Signalling, Railway, [28](#), [37](#), [47](#)
- Software Development Process, [10](#)
- Software vs. Data, [14](#), [15](#)
- Sonar, [40](#), [41](#)
- Southern Pacific Transportation Company, [54](#)
- Soyuz, [26](#), [36](#), [37](#)
- Staffing Data, [35](#)
- Submarine, [40](#), [41](#)
- Suppression Property, [8](#), [9](#), [27](#), [45](#)

Timeliness Property, [4](#), [6](#), [8](#), [9](#), [25–29](#), [31](#), [35](#), [42](#),
[44](#), [45](#), [50](#), [51](#), [54](#), [63](#)
Torque Calibration, [40](#)
Traceability Property, [8](#), [9](#), [25](#), [32](#), [63](#)
Training
 AI, [4](#), [7](#)
 Personnel, [35](#), [48](#), [53](#)
Trawler Karen, [26](#), [40](#), [41](#)
Turkish Airlines, [41](#)

Unknown Unknowns, [20](#)
Verifiability Property, [8](#), [9](#), [25–29](#), [34](#), [36](#), [37](#),
[42–44](#), [46](#), [48](#), [50](#), [52](#), [54](#)
Washington State Prison, [52](#)
What3Words, [25](#)
Zero-day, [25](#), [33](#)

DATA IS HERE. DATA IS GROWING. DATA IS CAUSING HARM.

This book has been developed by the Safety-Critical Systems Club Data Safety Initiative Working Group (DSIWG) to provide guidance on how data, as distinct from hardware and software can be managed in a safety-related context.

"If you torture the data long enough, they confess – even to crimes that were never committed."

Nihat Bülent Gültekin

~

This is the seventh minor update since version 3.0. Paragraph numbering within the body of the document remains aligned with that major release. Thus users of any previous 3.x release of the guidance document will find migration to this edition takes little effort.

